



数电实验

数字逻辑电路实验

- 实验六 可编程数字逻辑设计-内容1

李红艳

yanhongli523@seu.edu.cn



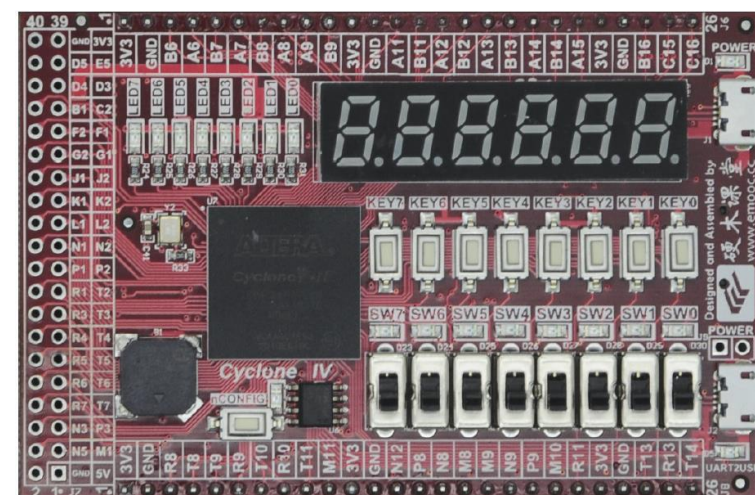
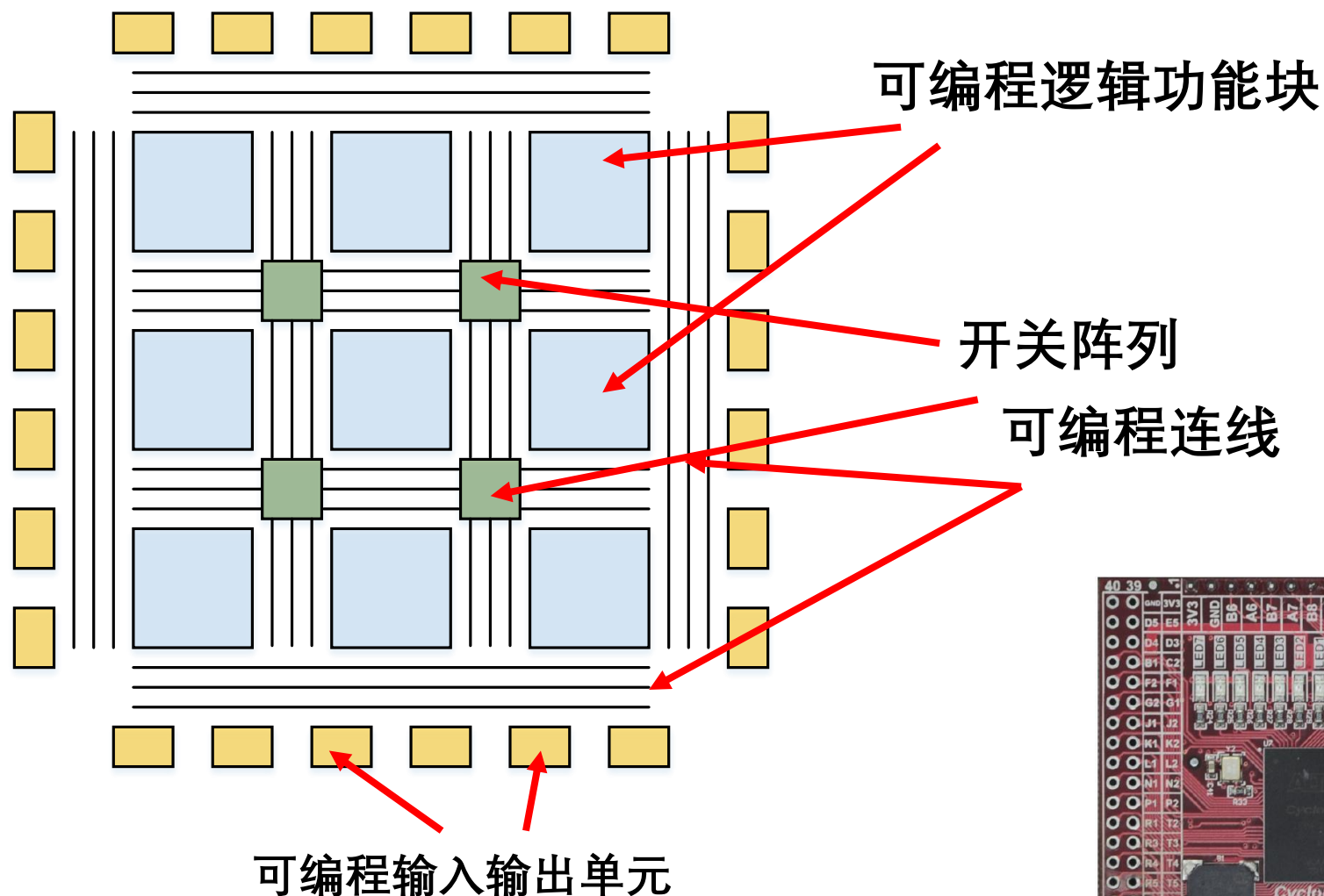
东南大学电工电子实验中心

实验目的

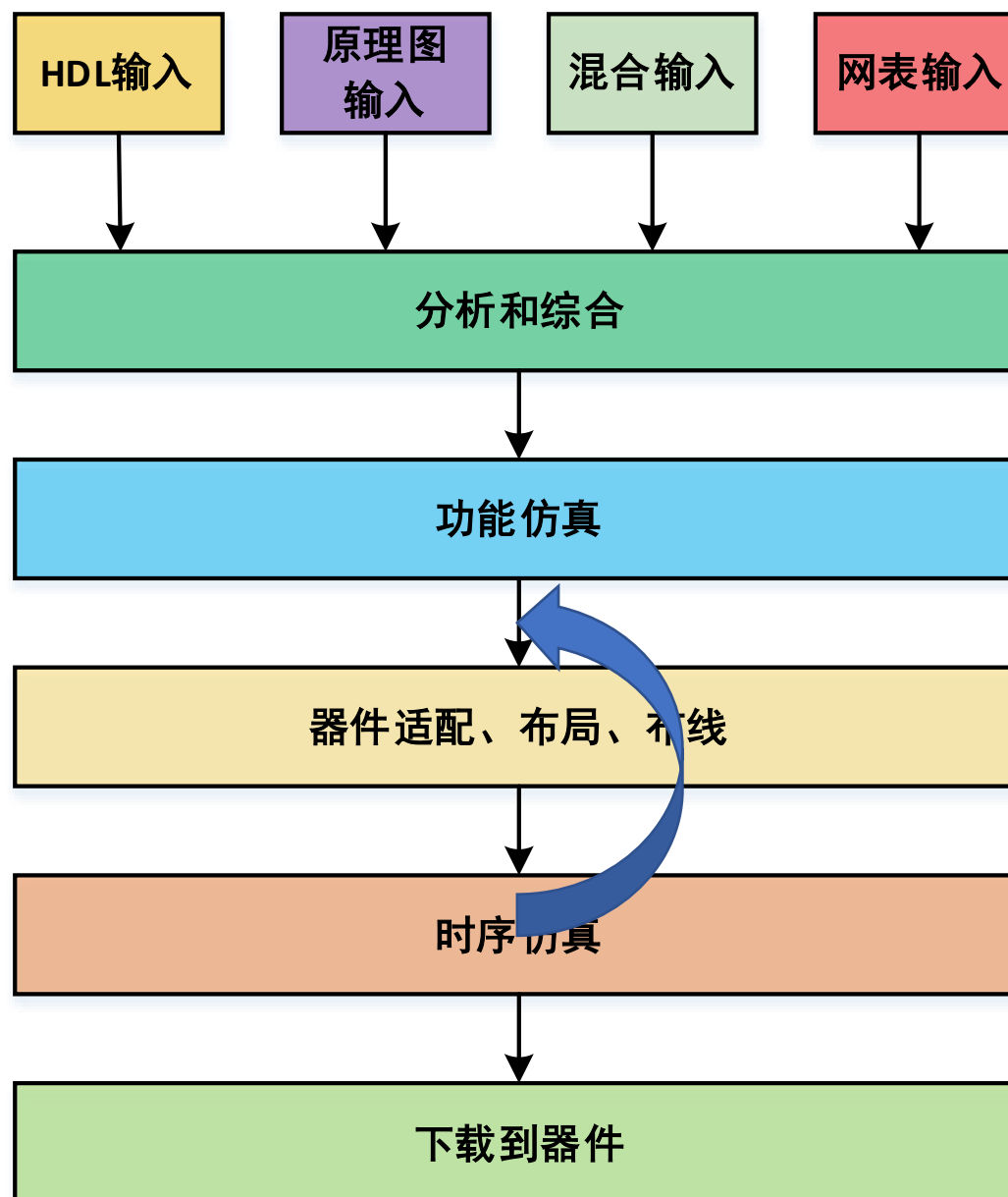
- 了解可编程数字系统设计的流程
- 掌握Quartus 软件的使用方法



FPGA基础内部结构



可编程数字逻辑设计流程





实验内容：模60同步计数器

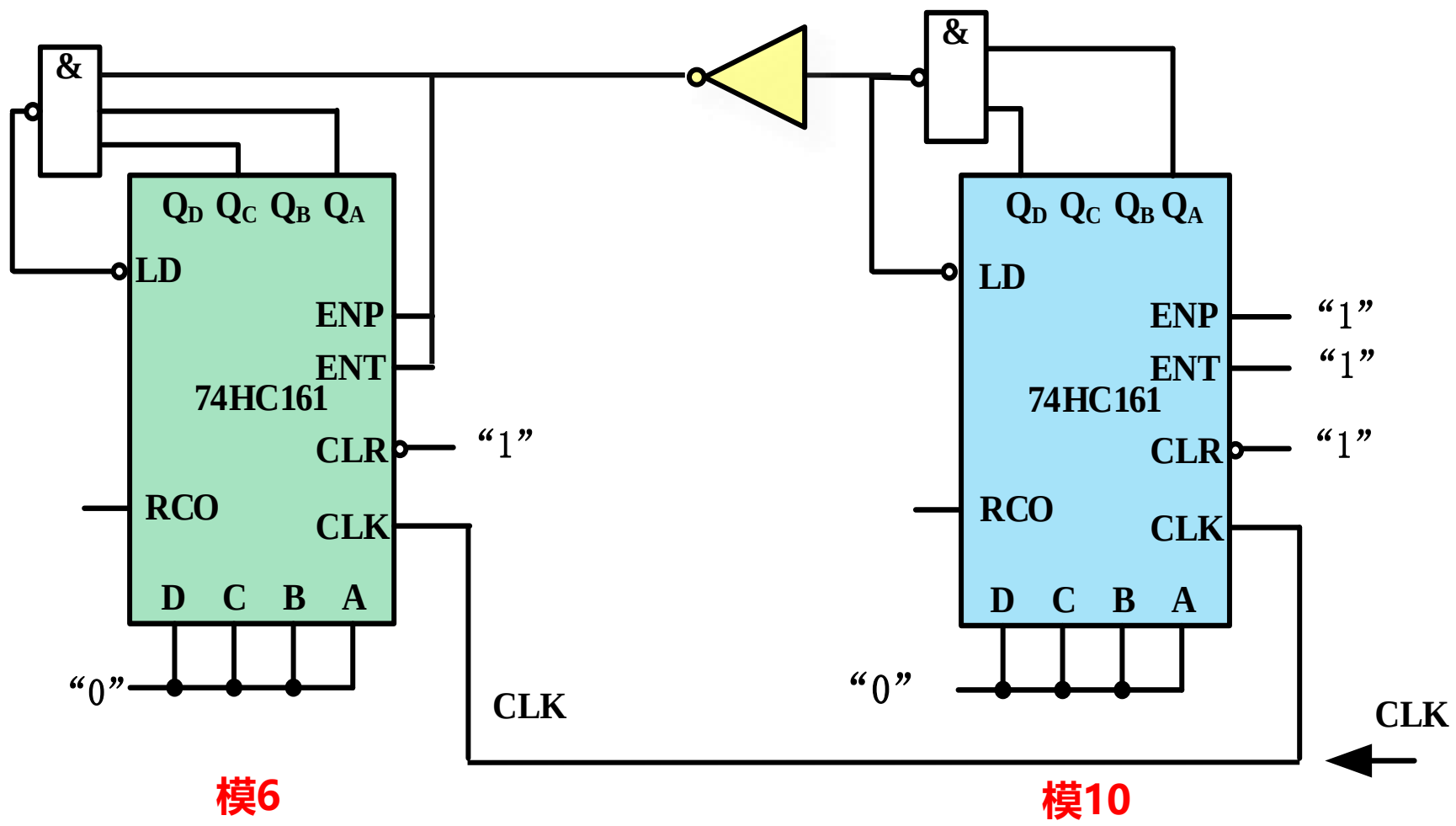
- ◆ 用74161设计一个模10同步计数器，并封装成元件
- ◆ 用74161设计一个模6同步计数器，并封装成元件
- ◆ 调用模6和模10计数器模块实现一个模60同步计数器
- ◆ 对模60进行功能和时序仿真验证
- ◆ 最后下载到实验箱进行实物验证



模60计数器

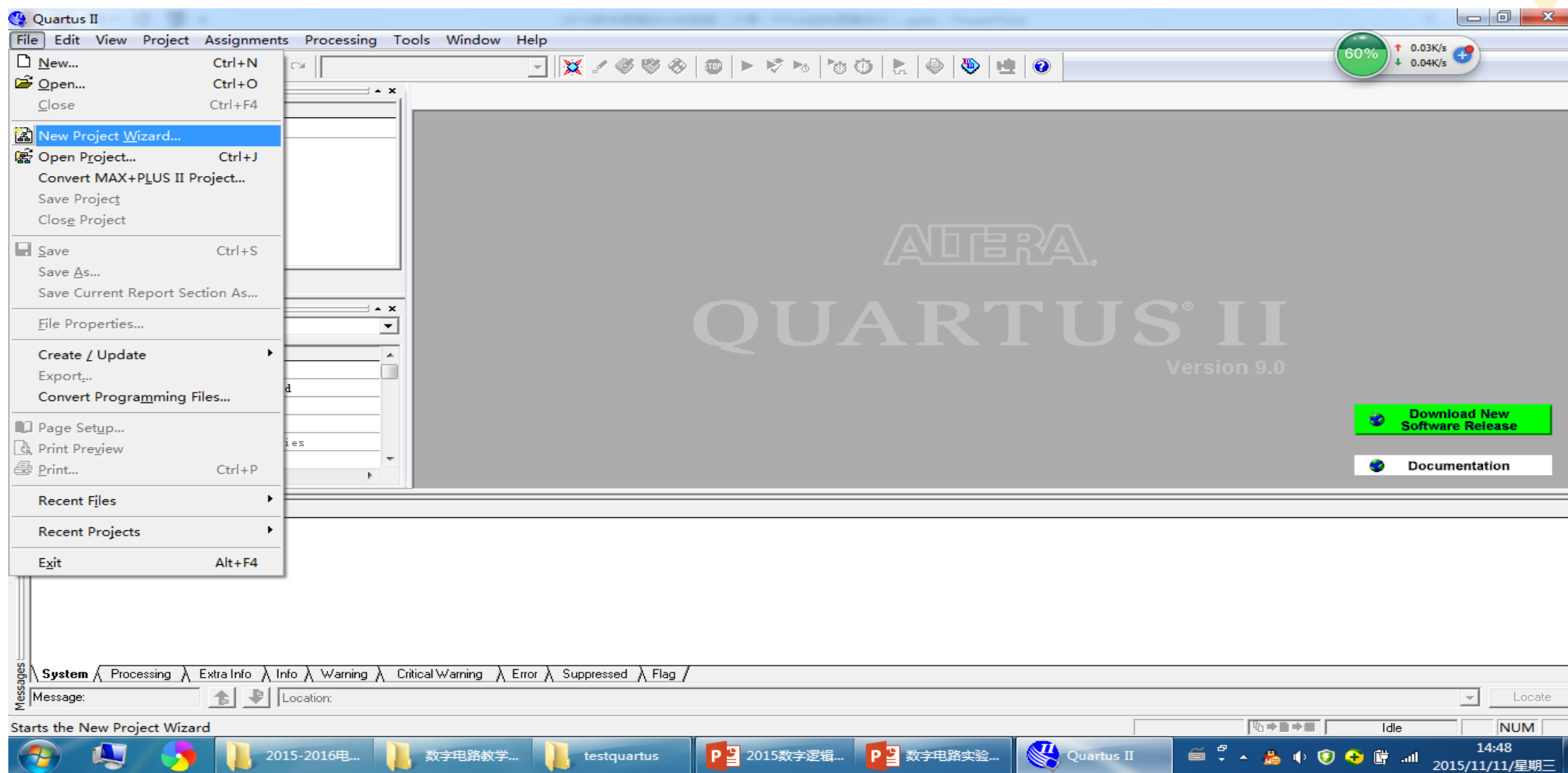
计到59置0

逢十进一
模10计数器置0

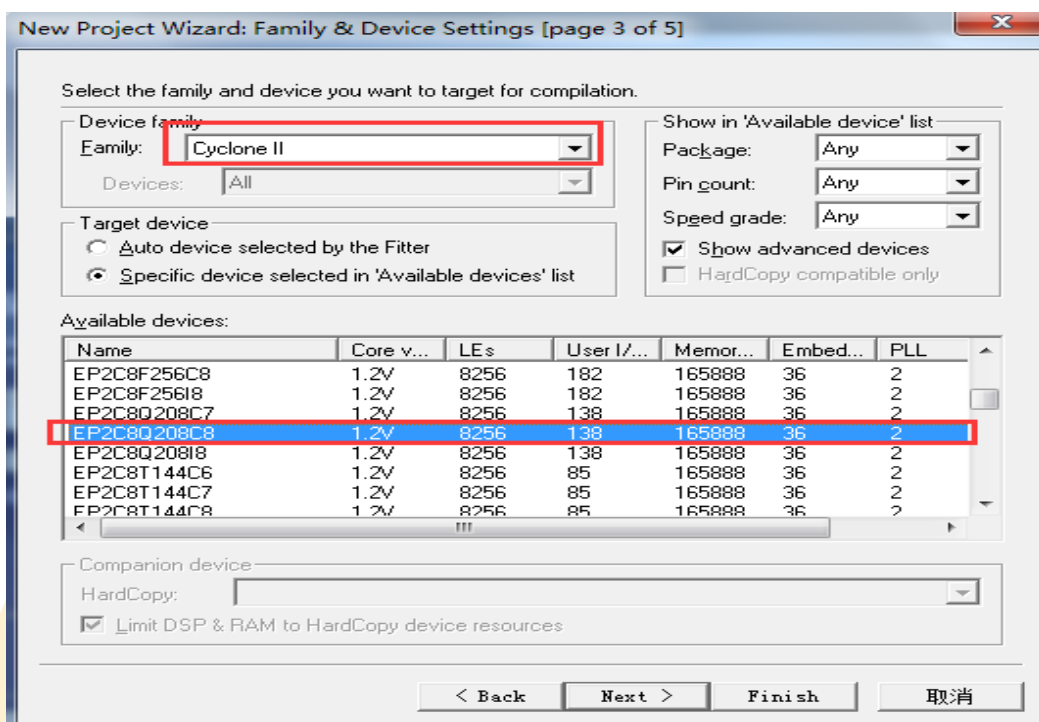
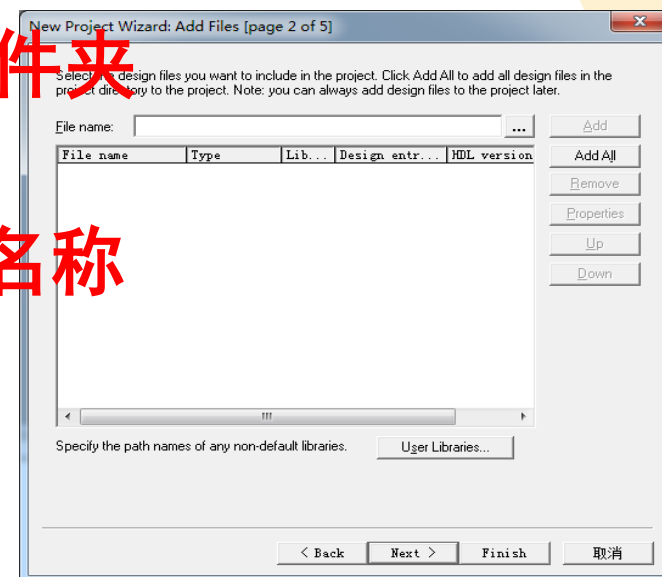
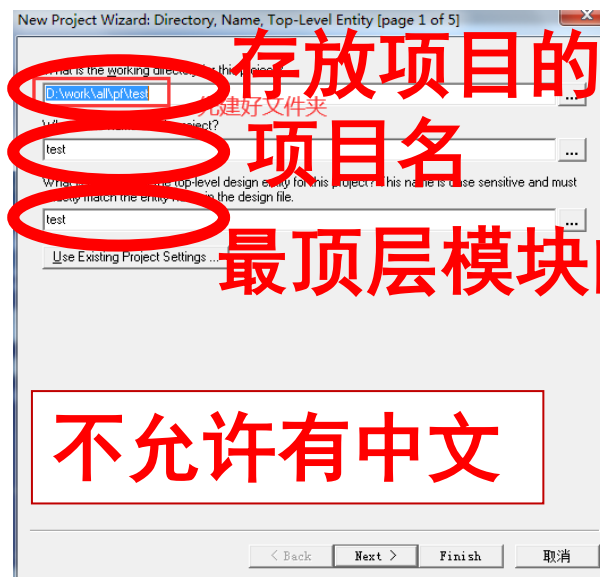
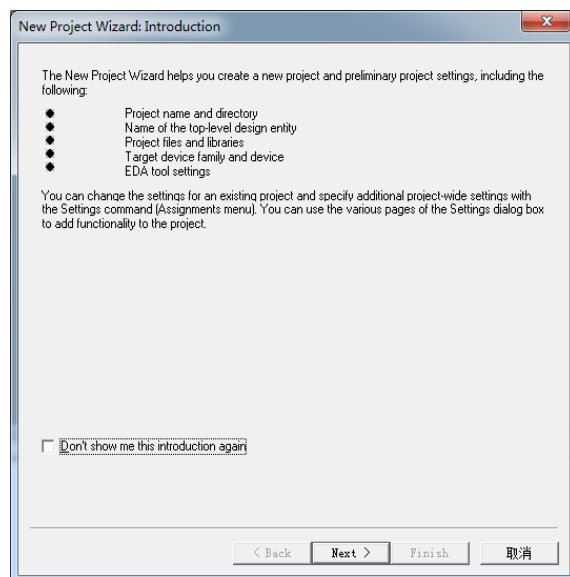


QuartusII软件的使用— Step1 项目创建

◆ File -> New Project Wizard



QuartusII软件的使用— Step1 项目创建



FPGA为EP4CE6F17C8N，但Quartus9.1sp2Web板不支持该器件的功能和时序仿真。
故项目创建Device先选择CycloneIII EP3C5E144C8。
时序仿真通过后再将器件切换回来。



QuartusII软件的使用— Step1 项目创建

New Project Wizard: EDA Tool Settings [page 4 of 5]

Specify the other EDA tools -- in addition to the Quartus II software -- used with the project.

Design Entry/Synthesis

Tool name:

Format:

☐ Run this tool automatically to synthesize the current design

Simulation

Tool name:

Format:

☐ Run gate-level simulation automatically after compilation

Timing Analysis

Tool name:

Format:

☐ Run this tool automatically after compilation

< Back Next > Finish 取消

New Project Wizard: Summary [page 5 of 5]

When you click Finish, the project will be created with the following settings:

Project directory:
D:/Quartus test/trafficLight/

Project name: trafficLight

Top-level design entity: trafficLight

Number of files added: 0

Number of user libraries added: 0

Device assignments:

Family name: Cyclone II

Device: EP2C8Q208C8

EDA tools:

Design entry/synthesis: <None>

Simulation: <None>

Timing analysis: <None>

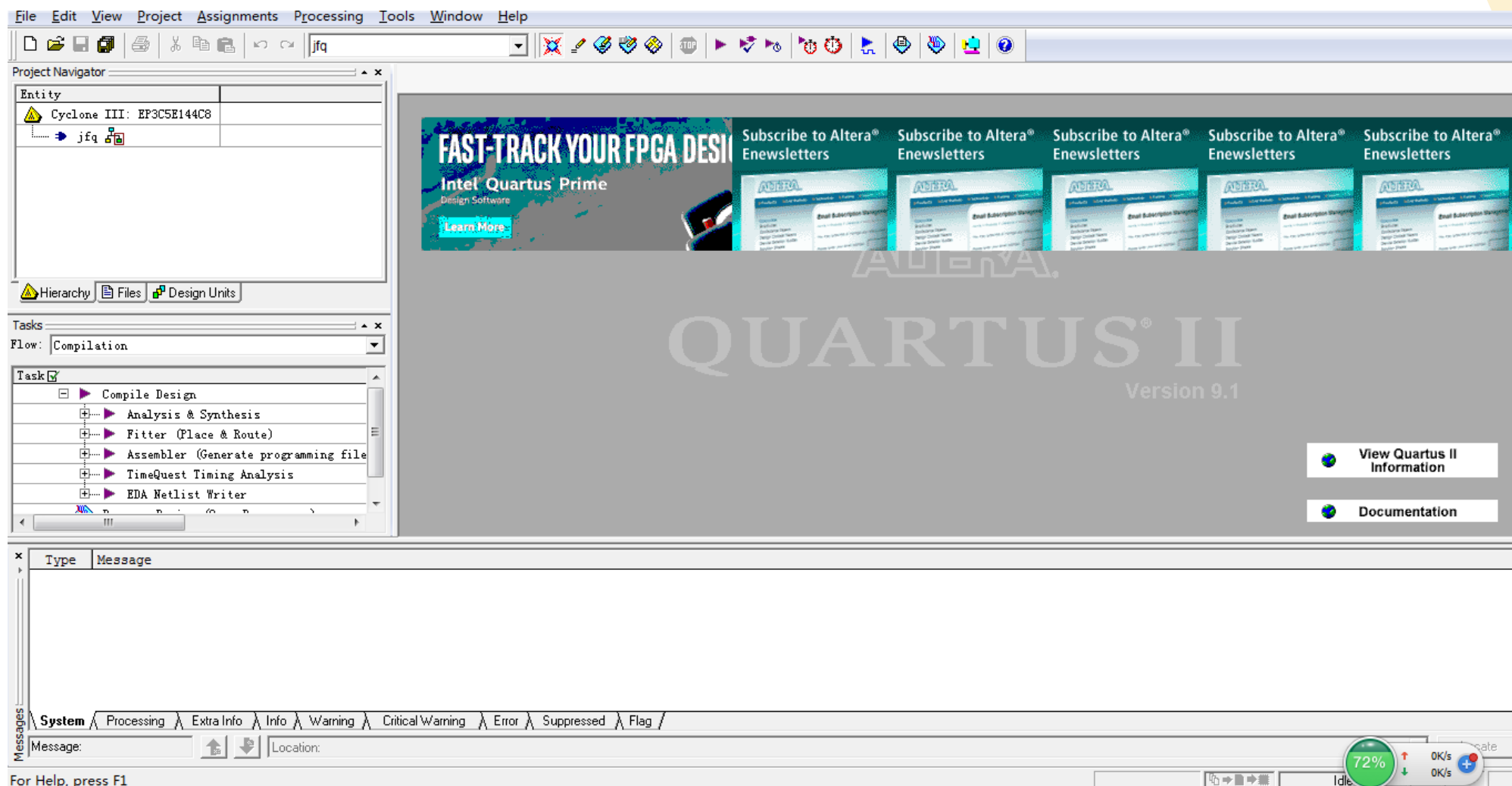
Operating conditions:

Core voltage: 1.2V

Junction temperature range: 0-85 癬

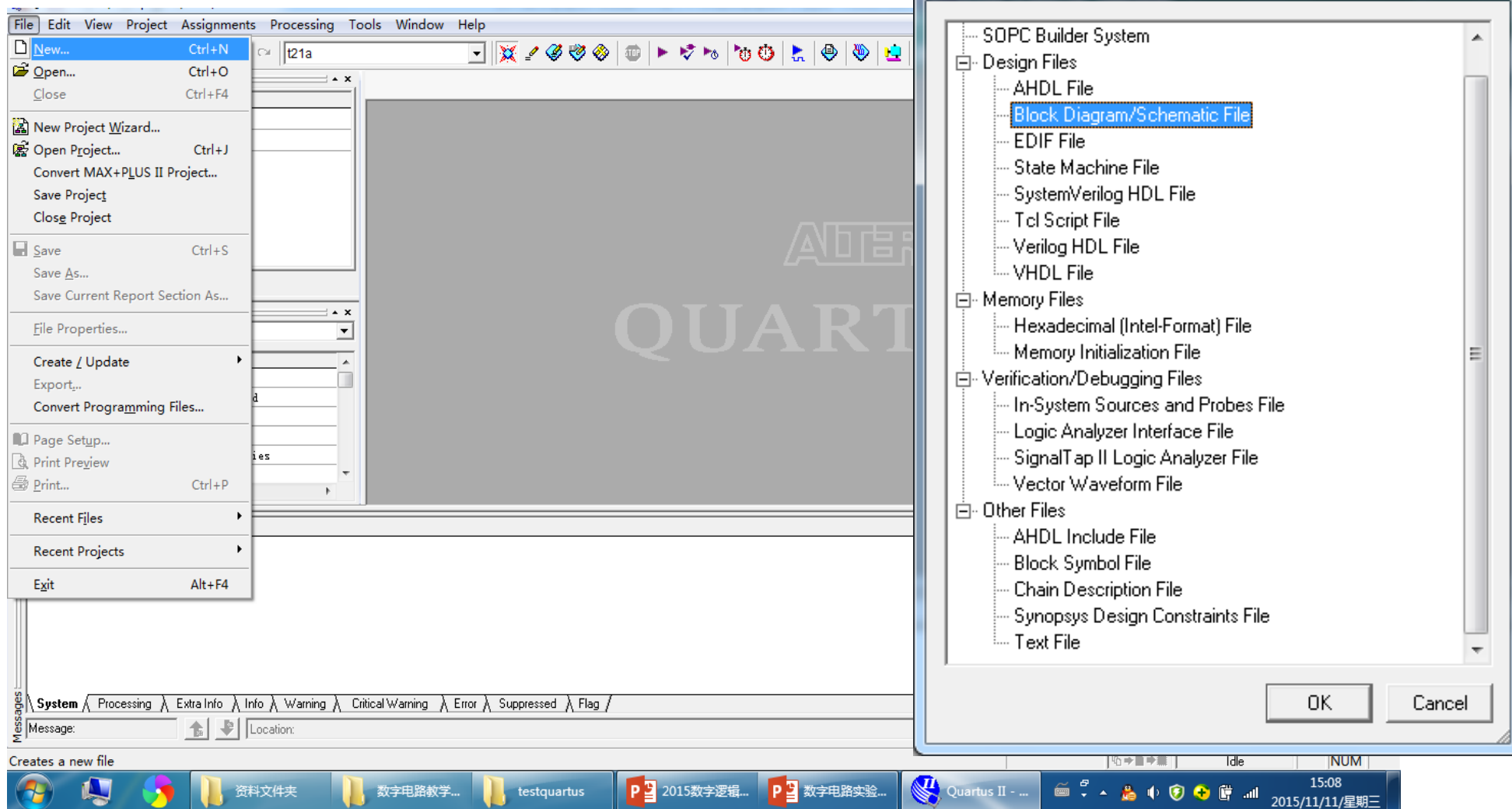
< Back Next > Finish 取消

QuartusII软件的使用— Step1 项目创建



QuartusII软件的使用— Step2 原理图输入

◆ 1、创建原理图文件

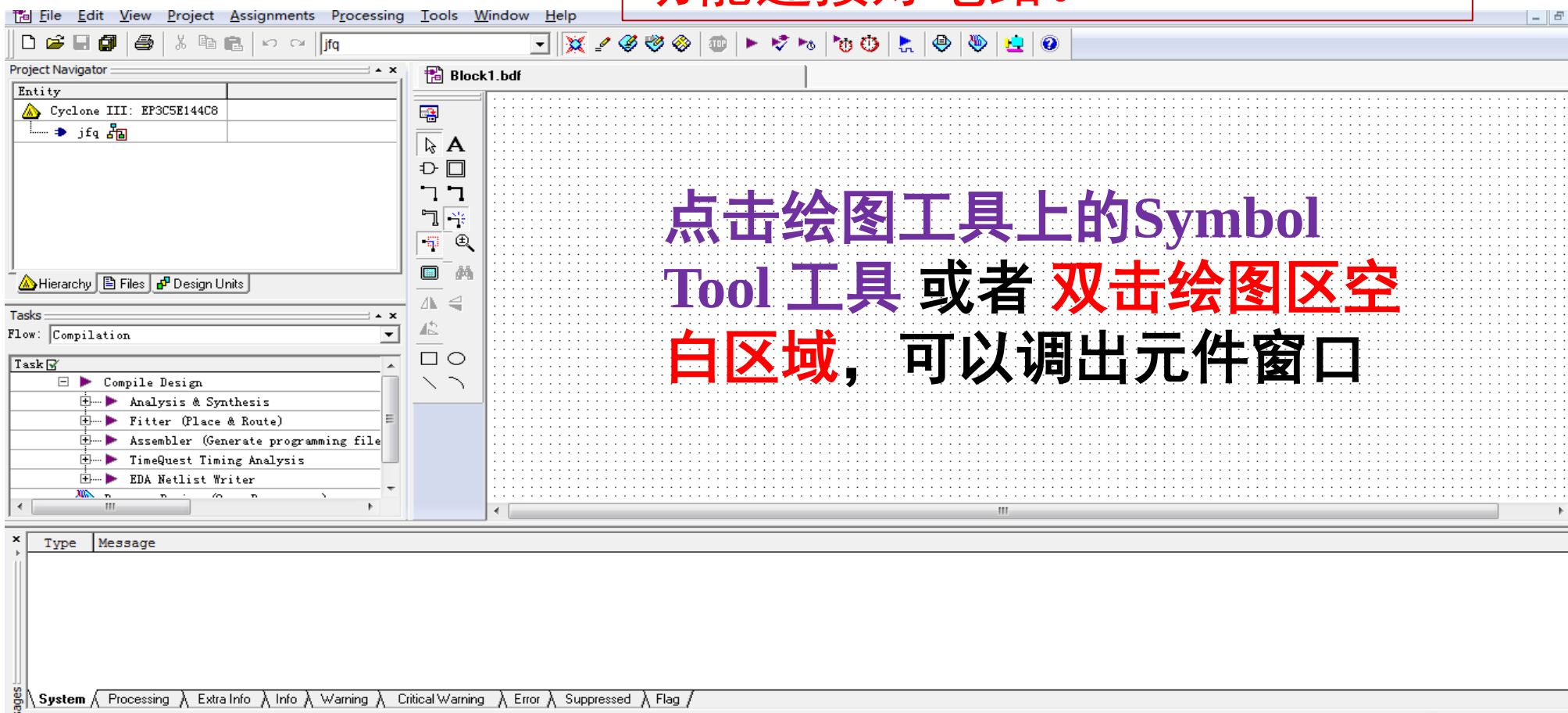




QuartusII软件的使用— Step2 原理图输入

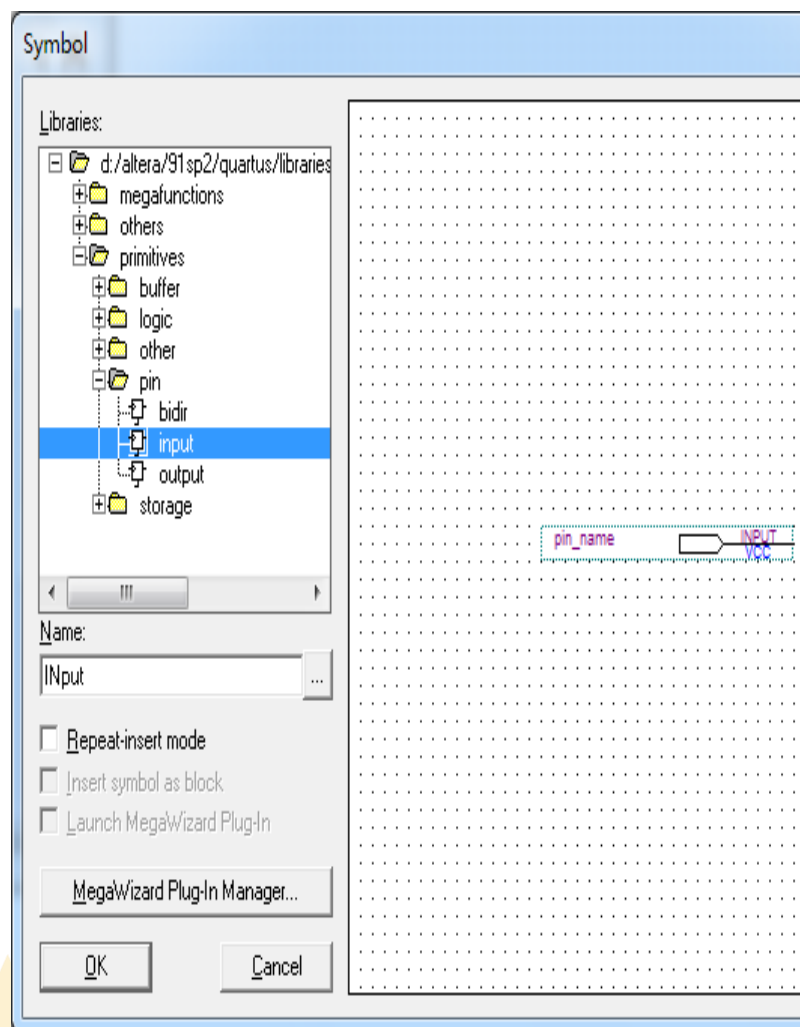
◆ 1、创建原理图文件

创建Block原理图文件，图中须定义I/O项、逻辑器件，并根据功能连接好电路。

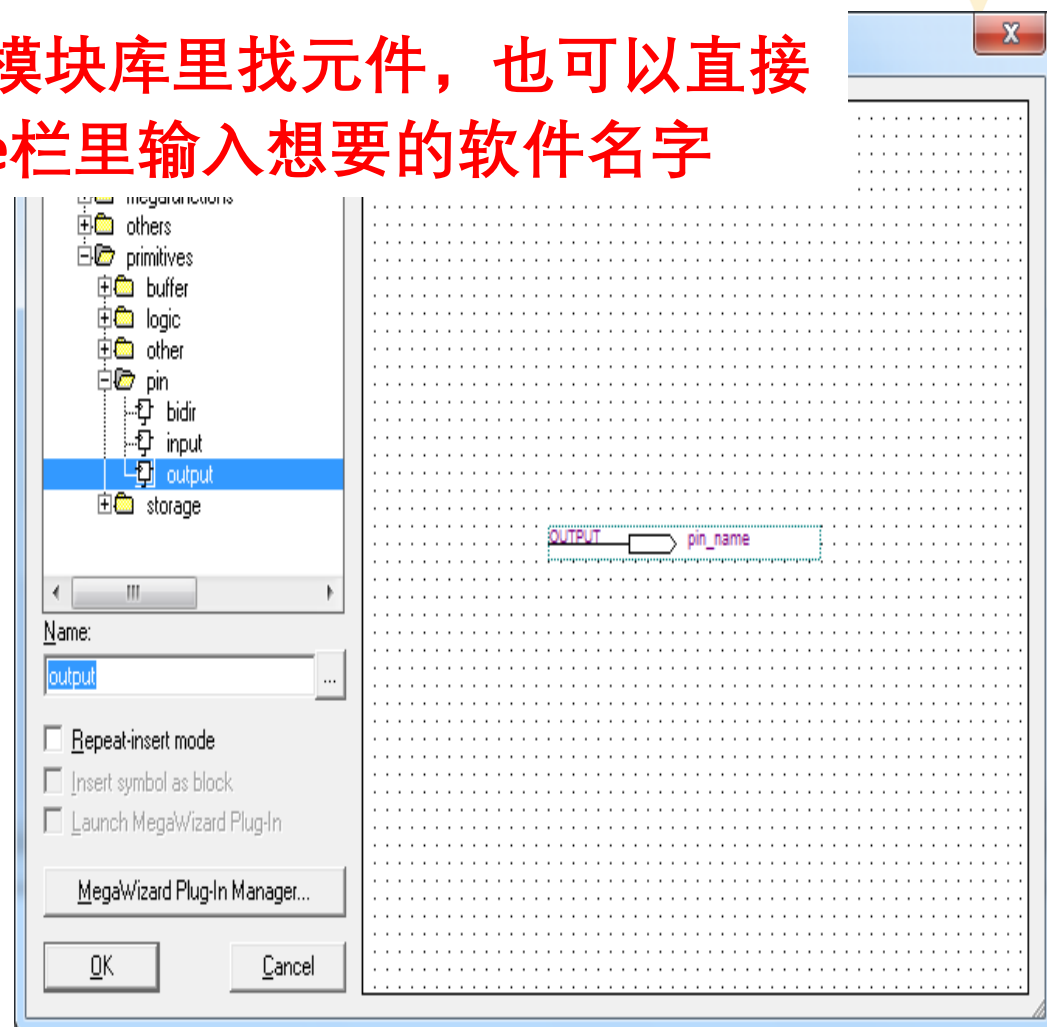


QuartusII软件的使用— Step2 原理图输入

◆ 2、元件输入及连接



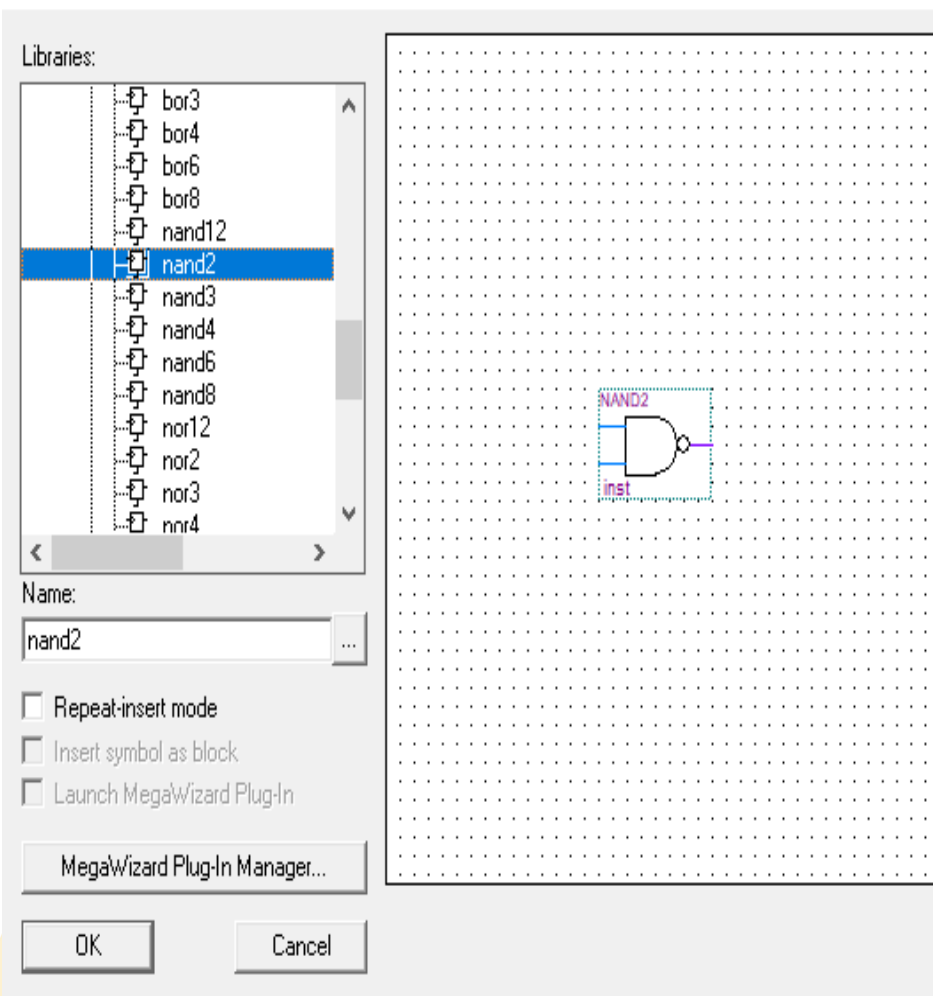
可以在模块库里找元件，也可以直接在Name栏里输入想要的软件名字



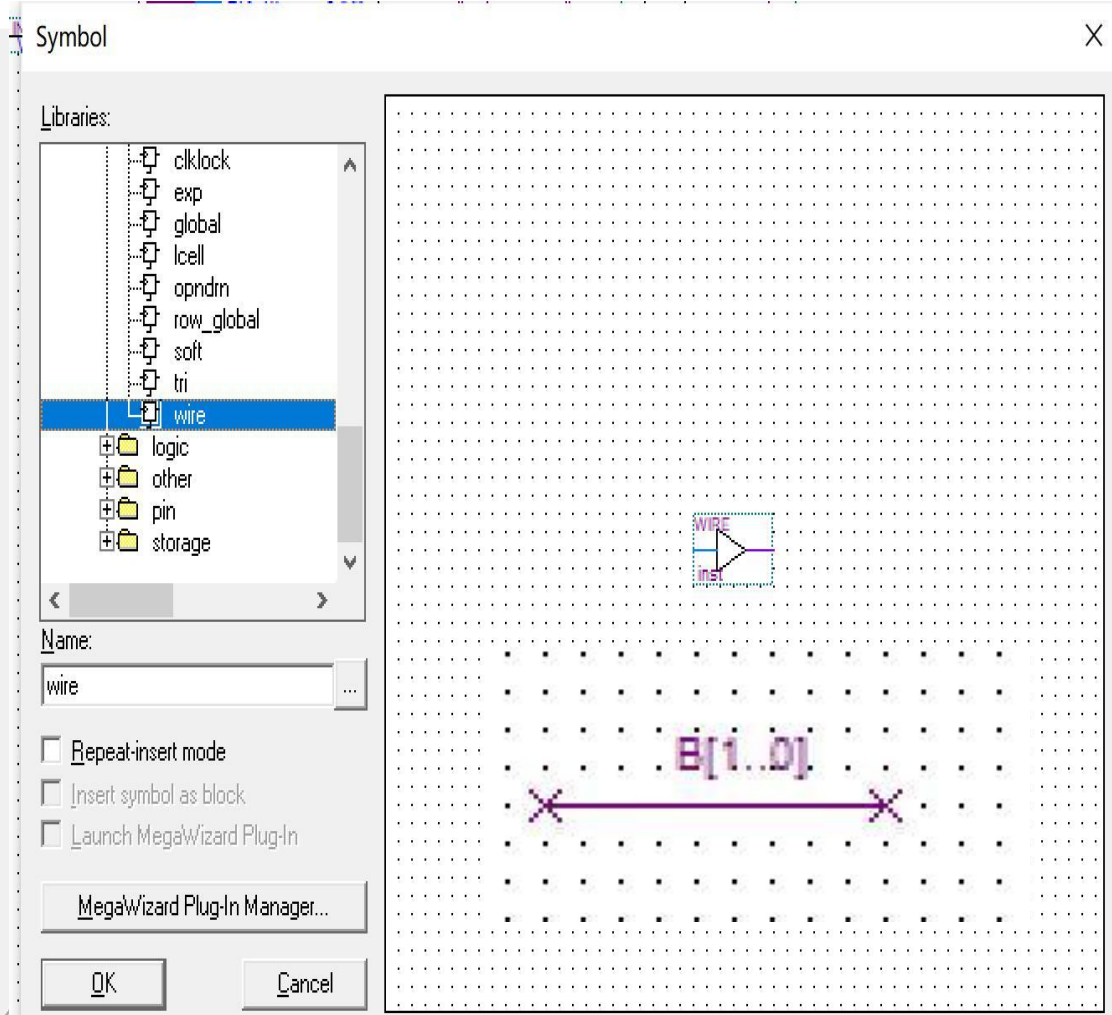
QuartusII软件的使用— Step2 原理图输入

◆ 2、元件输入及连接

Symbol



X

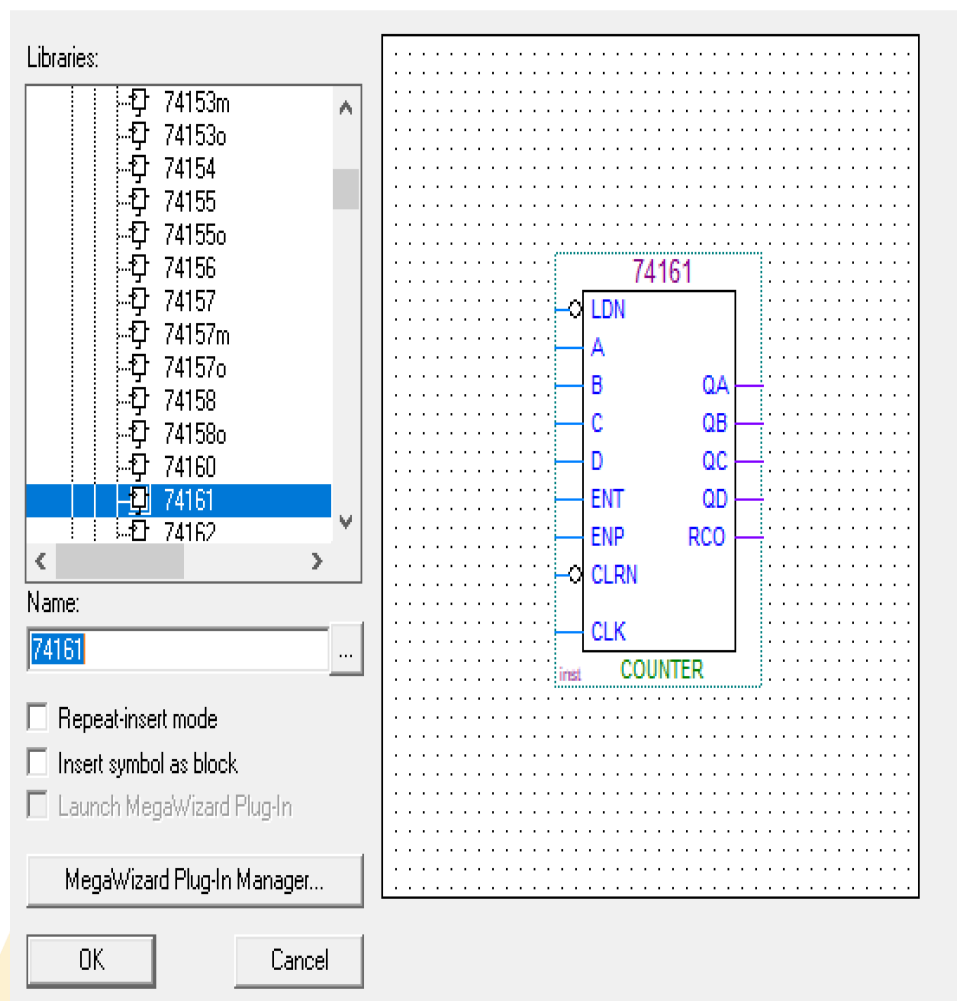


QuartusII软件的使用— Step2 原理图输入

◆ 2. 元件输入及连接

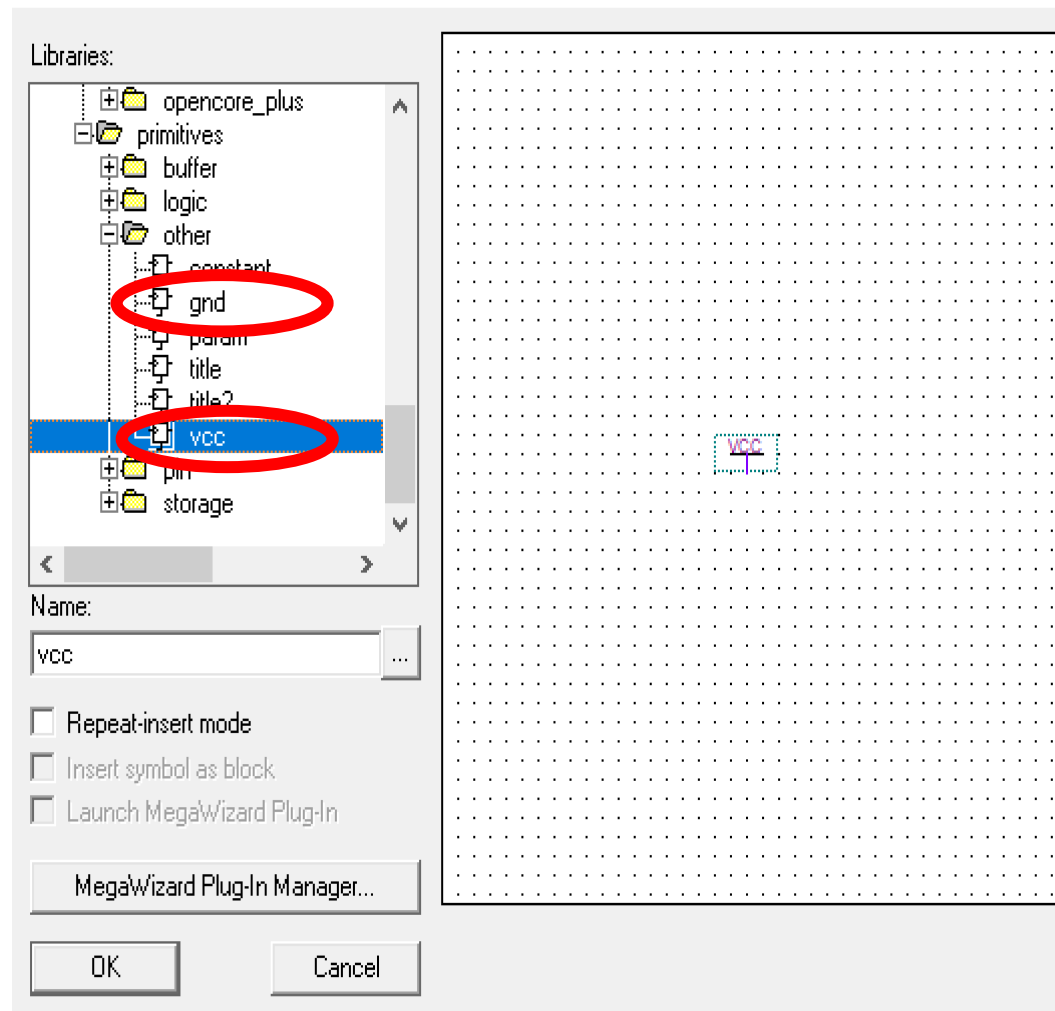
Symbol

X



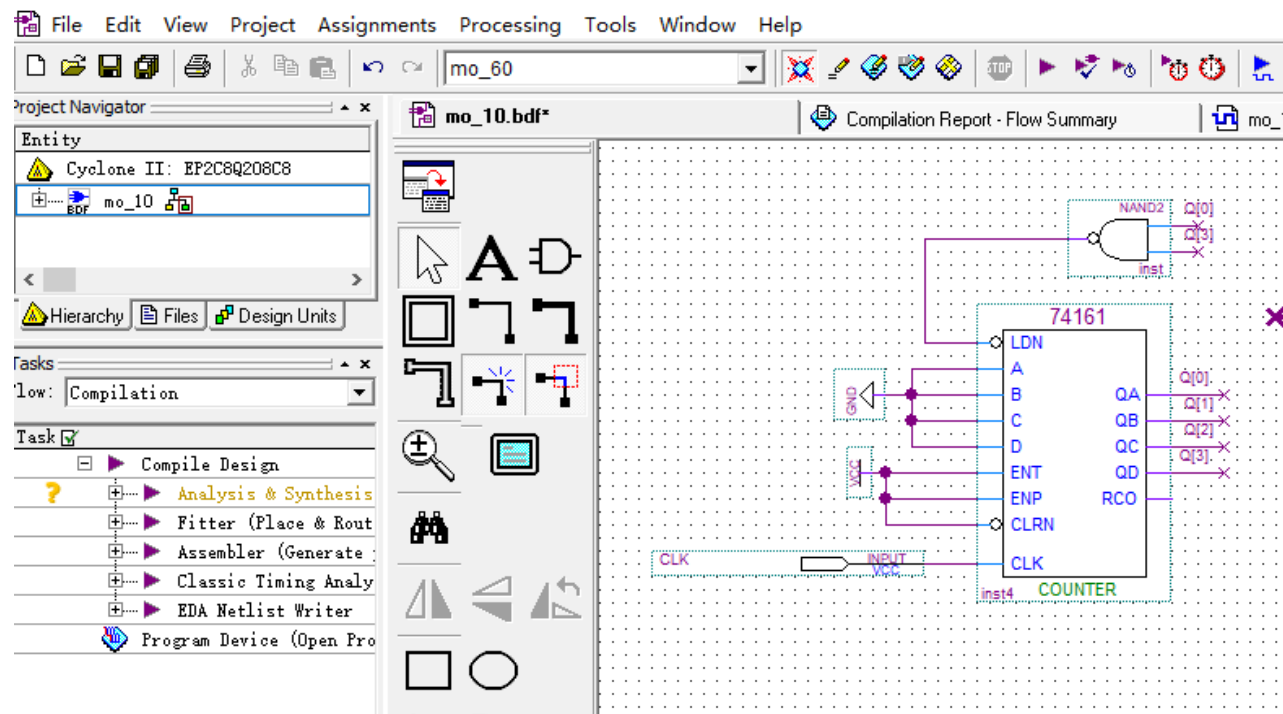
Symbol

X



QuartusII软件的使用— Step2 原理图输入

◆ 2、元件输入及连接



注意下命名格式：
总线：Q[3..0]，
bit位：Q[3]，Q[2]，
Q[1]，Q[0]

命名相同的总线和多位单线，电气连接相通。
Bus总线可整体作运算，
也可分bit位作运算

mo_60 -c mo_60

s is enabled

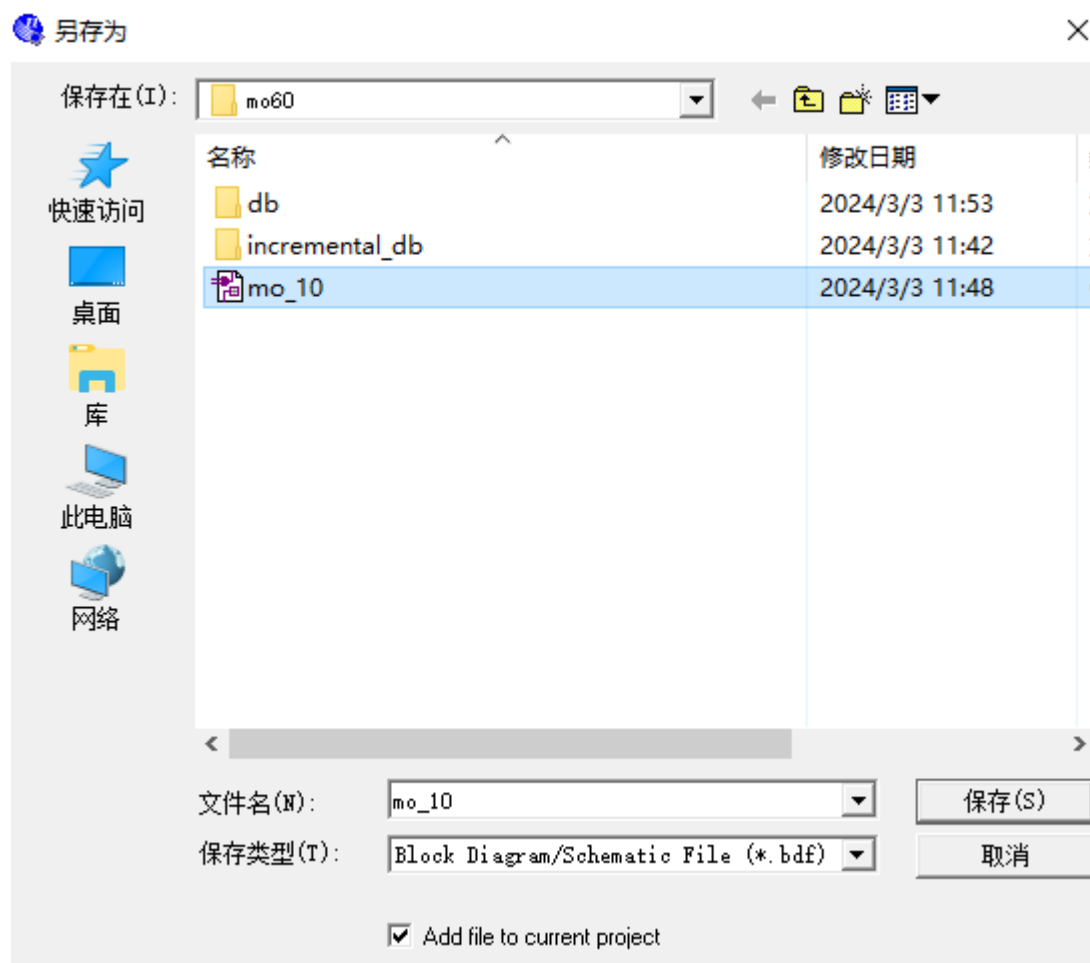
nto CVWF format in order to reduce file size. For more details please refer to the

lag /

Locate

QuartusII软件的使用— Step2 原理图输入

◆ 3、文件保存



QuartusII软件的使用—Step3 项目处理

◆分析(Analysis)

- 检查工程的逻辑完整性和一致性，并检查边界连接和语法错误

◆综合(Synthesis)

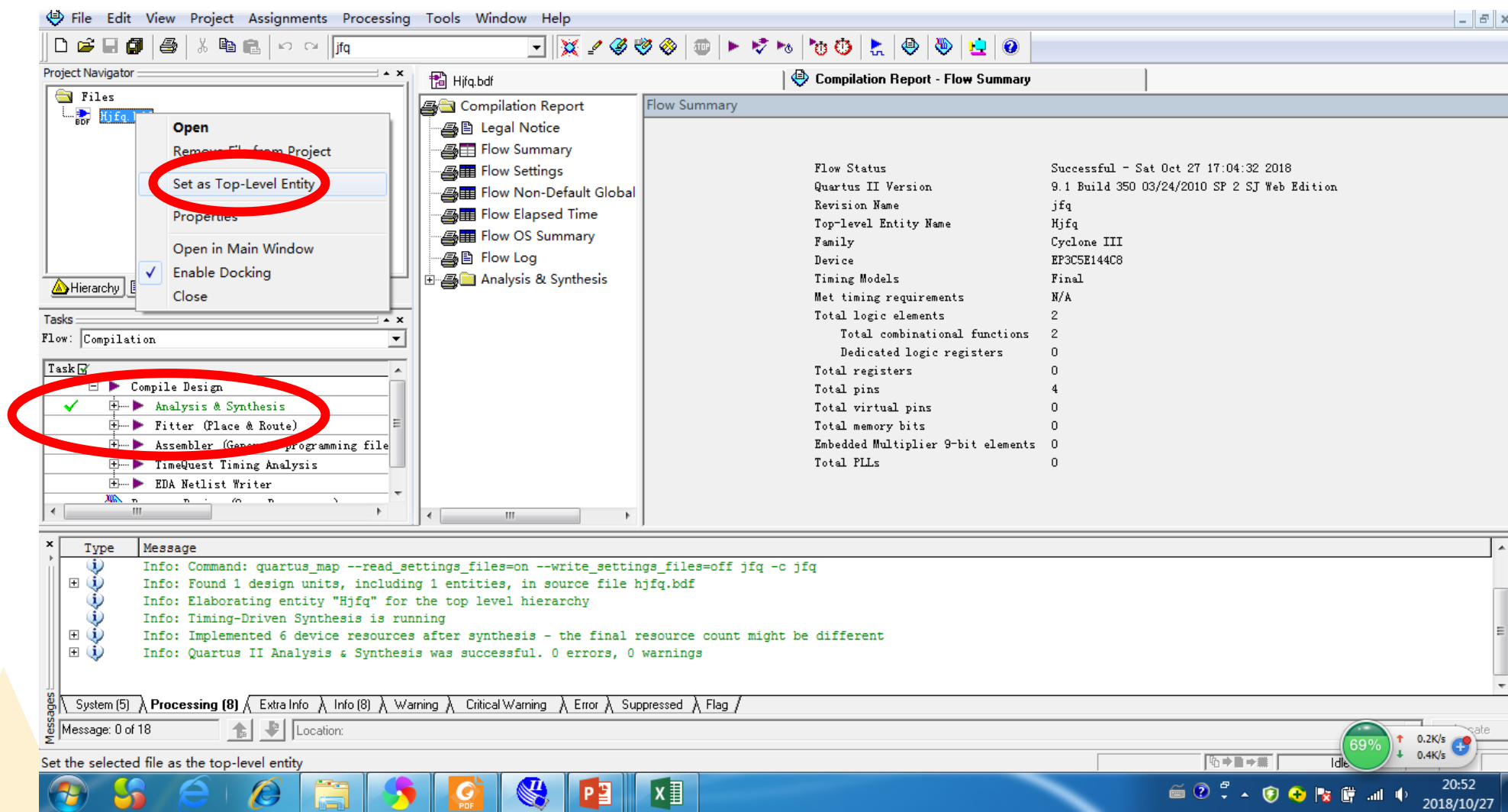
- 在设计实体或工程文件的逻辑上进行综合和技术映射，
- 综合器可以从HDL中推断触发器、锁存器和状态机
- 为状态机建立状态分配，并作出能减少所用资源的选择
- 使用多种算法来减少门的数量，删除冗余逻辑实现时序要求
- 优化设计以尽可能有效地利用器件体系结构

将当前.bdf文件设为顶层文件，点击主工具栏上的  按钮，开始 “Analysis and Synthesis”编译过程

Quartus II软件的使用— Step3项目处理

◆ 1、顶层文件设置

将需要仿真的文件设置为顶层文件



The screenshot displays the Quartus II software interface during a project setup. The **Project Navigator** on the left shows a file named **Hjfq.bdf**. A right-click context menu is open over this file, with the option **Set as Top-Level Entity** highlighted by a red circle. Below the menu, the **Tasks** pane shows the **Compile Design** task selected, with its sub-tasks listed: **Analysis & Synthesis** (checked), **Fitter (Place & Route)**, **Assembler (Generate programming file)**, **TimeQuest Timing Analysis**, and **EDA Netlist Writer**. The **Compilation Report - Flow Summary** window on the right provides details about the compilation process.

Flow Summary	
Flow Status	Successful - Sat Oct 27 17:04:32 2018
Quartus II Version	9.1 Build 350 03/24/2010 SP 2 SJ Web Edition
Revision Name	jfq
Top-level Entity Name	Hjq
Family	Cyclone III
Device	EP3C5E144C8
Timing Models	Final
Met timing requirements	N/A
Total logic elements	2
Total combinational functions	2
Dedicated logic registers	0
Total registers	0
Total pins	4
Total virtual pins	0
Total memory bits	0
Embedded Multiplier 9-bit elements	0
Total PLLs	0

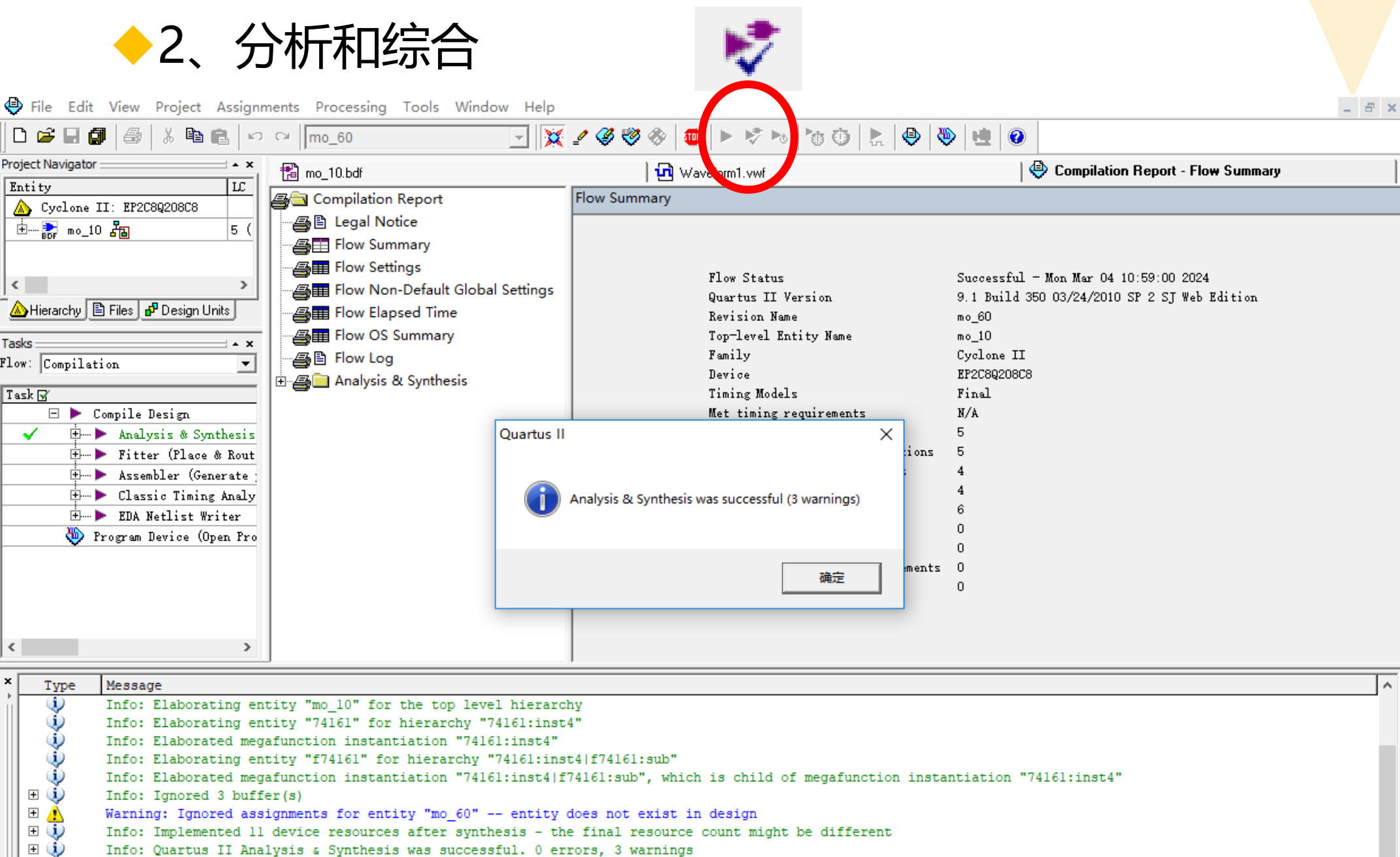
The **Messages** window at the bottom shows the following log:

```
Info: Command: quartus_map --read_settings_files=on --write_settings_files=off jq -c jq
Info: Found 1 design units, including 1 entities, in source file hjfq.bdf
Info: Elaborating entity "Hjq" for the top level hierarchy
Info: Timing-Driven Synthesis is running
Info: Implemented 6 device resources after synthesis - the final resource count might be different
Info: Quartus II Analysis & Synthesis was successful. 0 errors, 0 warnings
```

At the bottom of the interface, a status bar indicates the progress of the synthesis: **Set the selected file as the top-level entity** (69% complete).

Quartus II软件的使用— Step3项目处理

◆ 2、分析和综合



The image displays the Quartus II software interface during the Analysis & Synthesis process. The top toolbar features a red circle around the 'Analysis & Synthesis' button. The 'Flow Summary' window shows the process is successful. A dialog box confirms 'Analysis & Synthesis was successful (3 warnings)'. The bottom message window shows detailed logs.

Flow Summary

Flow Status	Successful - Mon Mar 04 10:59:00 2024
Quartus II Version	9.1 Build 350 03/24/2010 SP 2 SJ Web Edition
Revision Name	mo_60
Top-level Entity Name	mo_10
Family	Cyclone II
Device	EP2C8Q208C8
Timing Models	Final
Met timing requirements	N/A

Quartus II

Analysis & Synthesis was successful (3 warnings)

确定

Message

Type	Message
Info	Info: Elaborating entity "mo_10" for the top level hierarchy
Info	Info: Elaborating entity "74161" for hierarchy "74161:inst4"
Info	Info: Elaborated megafunction instantiation "74161:inst4"
Info	Info: Elaborating entity "f74161" for hierarchy "74161:inst4 f74161:sub"
Info	Info: Elaborated megafunction instantiation "74161:inst4 f74161:sub", which is child of megafunction instantiation "74161:inst4"
Info	Info: Ignored 3 buffer(s)
Warning	Warning: Ignored assignments for entity "mo_60" -- entity does not exist in design
Info	Info: Implemented 11 device resources after synthesis - the final resource count might be different
Info	Info: Quartus II Analysis & Synthesis was successful. 0 errors, 3 warnings

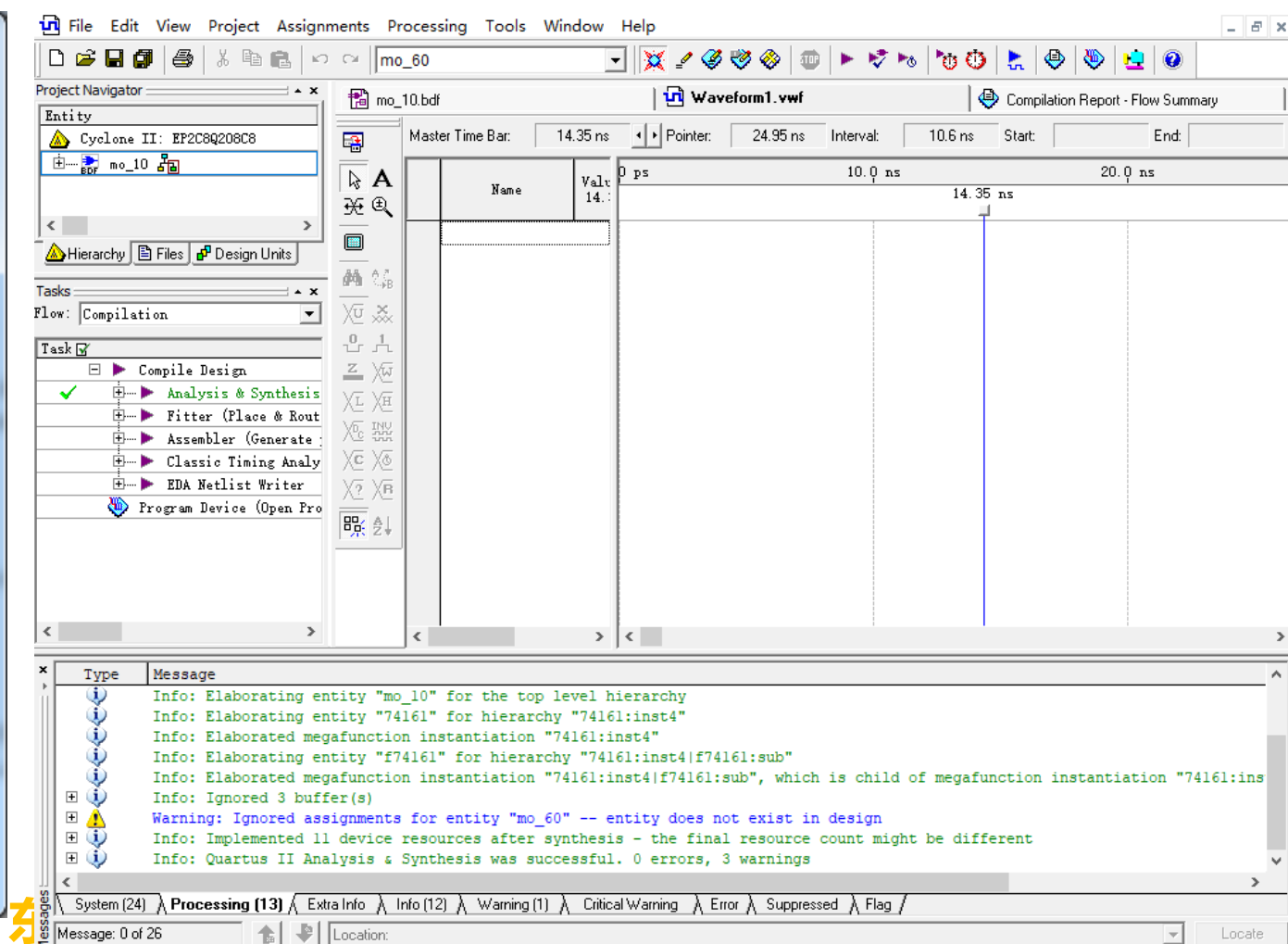
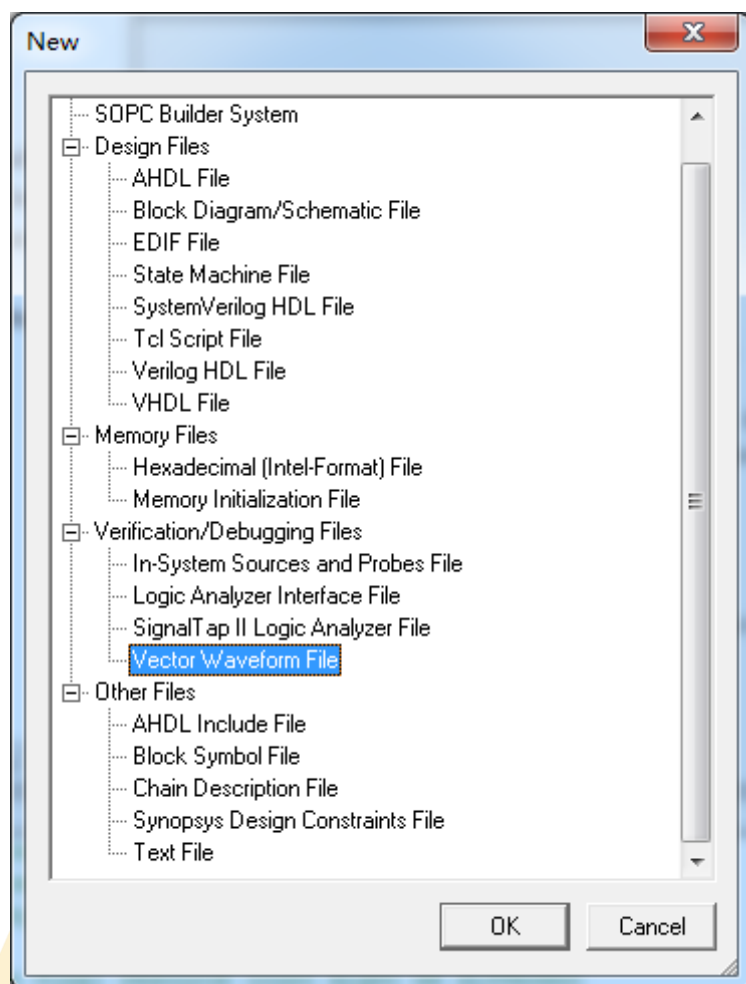


QuartusII软件的使用— Step4功能仿真

- ◆ 软件仿真，检验逻辑功能和内部定时是否符合设计要求
- ◆ 包括功能仿真和时序仿真
 - Processing->Simulator Tool
- ◆ 时序仿真需要计算大量的时延信息，仿真速度慢

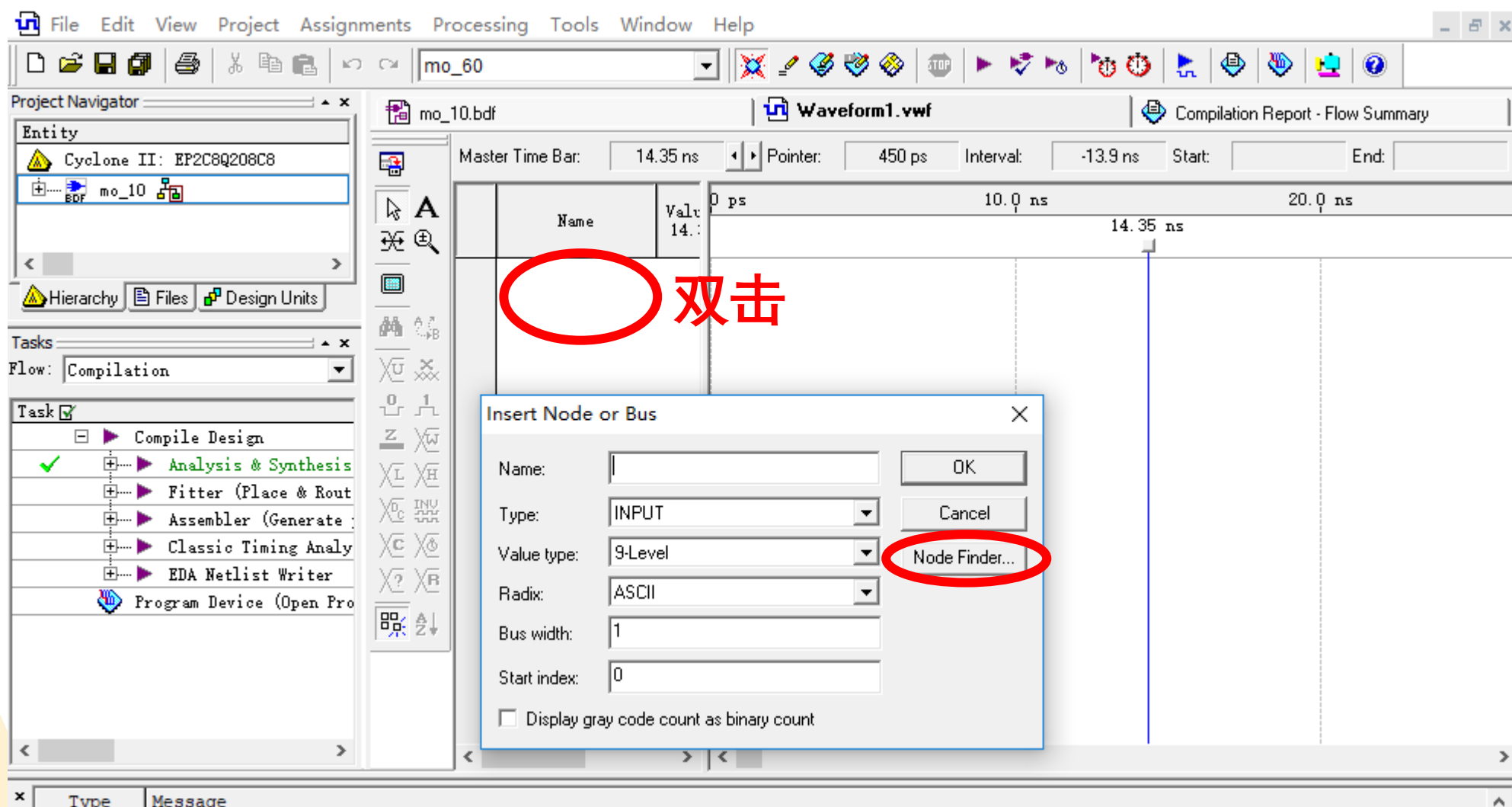
QuartusII软件的使用— Step4功能仿真

- ◆ 1、创建仿真文件 (.vwf)
 - File-> New -> New Waveform File



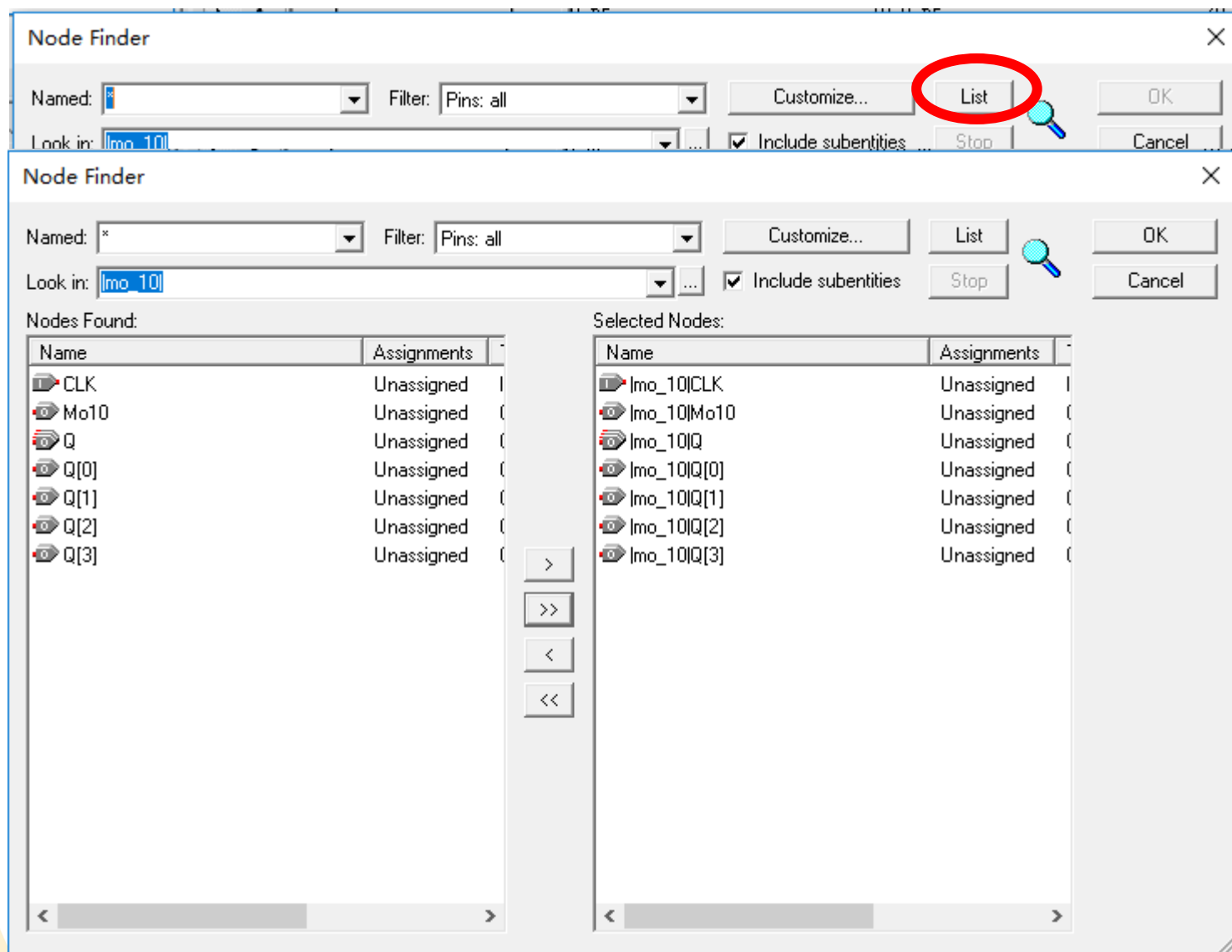
QuartusII软件的使用— Step4功能仿真

◆ 2、添加输入输出节点



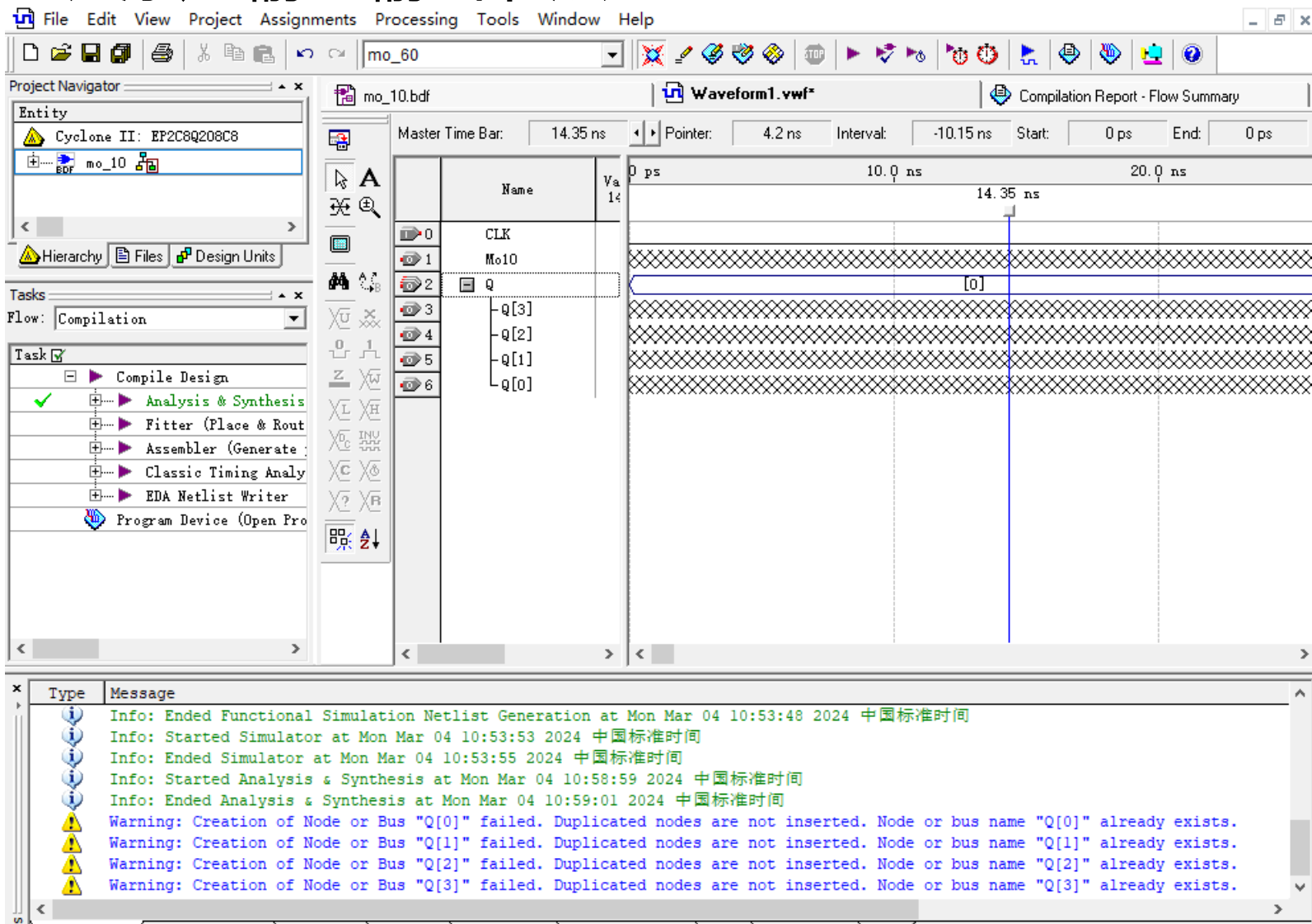
QuartusII软件的使用— Step4功能仿真

◆ 2、添加输入输出节点



QuartusII软件的使用— Step4功能仿真

◆ 2、添加输入输出节点



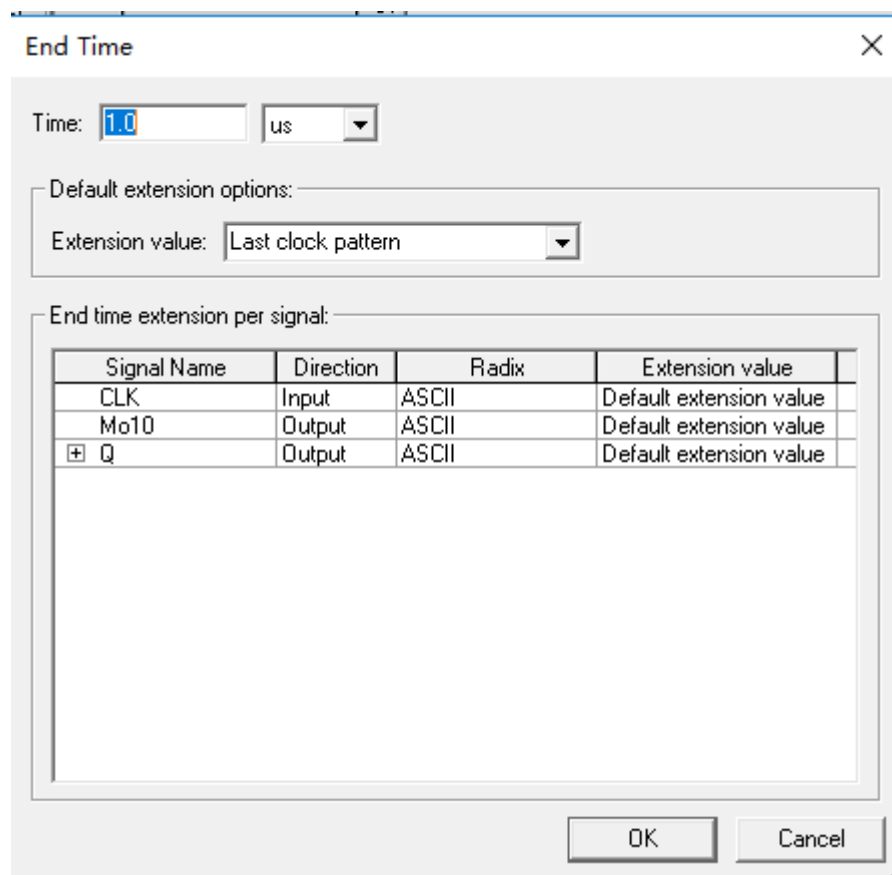
The screenshot displays the Quartus II software interface during a functional simulation. The top menu bar includes File, Edit, View, Project, Assignments, Processing, Tools, Window, and Help. The Project Navigator on the left shows the project hierarchy for 'Cyclone II: EP2C8Q208C8', with 'mo_10' selected. The Tasks window shows the 'Compile Design' task, with 'Analysis & Synthesis' completed. The main window displays a waveform for 'Waveform1.vwf' with a Master Time Bar at 14.35 ns. The waveform shows signals for CLK, Mo10, and Q[3], Q[2], Q[1], and Q[0]. The Q signals are shown as a single horizontal line at 0 ps, indicating they are not yet defined. The Message window at the bottom shows the following messages:

- Info: Ended Functional Simulation Netlist Generation at Mon Mar 04 10:53:48 2024 中国标准时间
- Info: Started Simulator at Mon Mar 04 10:53:53 2024 中国标准时间
- Info: Ended Simulator at Mon Mar 04 10:53:55 2024 中国标准时间
- Info: Started Analysis & Synthesis at Mon Mar 04 10:58:59 2024 中国标准时间
- Info: Ended Analysis & Synthesis at Mon Mar 04 10:59:01 2024 中国标准时间
- Warning: Creation of Node or Bus "Q[0]" failed. Duplicated nodes are not inserted. Node or bus name "Q[0]" already exists.
- Warning: Creation of Node or Bus "Q[1]" failed. Duplicated nodes are not inserted. Node or bus name "Q[1]" already exists.
- Warning: Creation of Node or Bus "Q[2]" failed. Duplicated nodes are not inserted. Node or bus name "Q[2]" already exists.
- Warning: Creation of Node or Bus "Q[3]" failed. Duplicated nodes are not inserted. Node or bus name "Q[3]" already exists.

QuartusII软件的使用— Step4功能仿真

◆ 3、为输入信号建立激励波形

- Edit -> End Time, 设置仿真终止时间



The dialog box titled "End Time" is used to set the simulation end time. It includes a "Time" field set to "1.0" with a unit dropdown set to "us". Below this, there are "Default extension options" and "End time extension per signal" sections.

Time:

Default extension options:

Extension value:

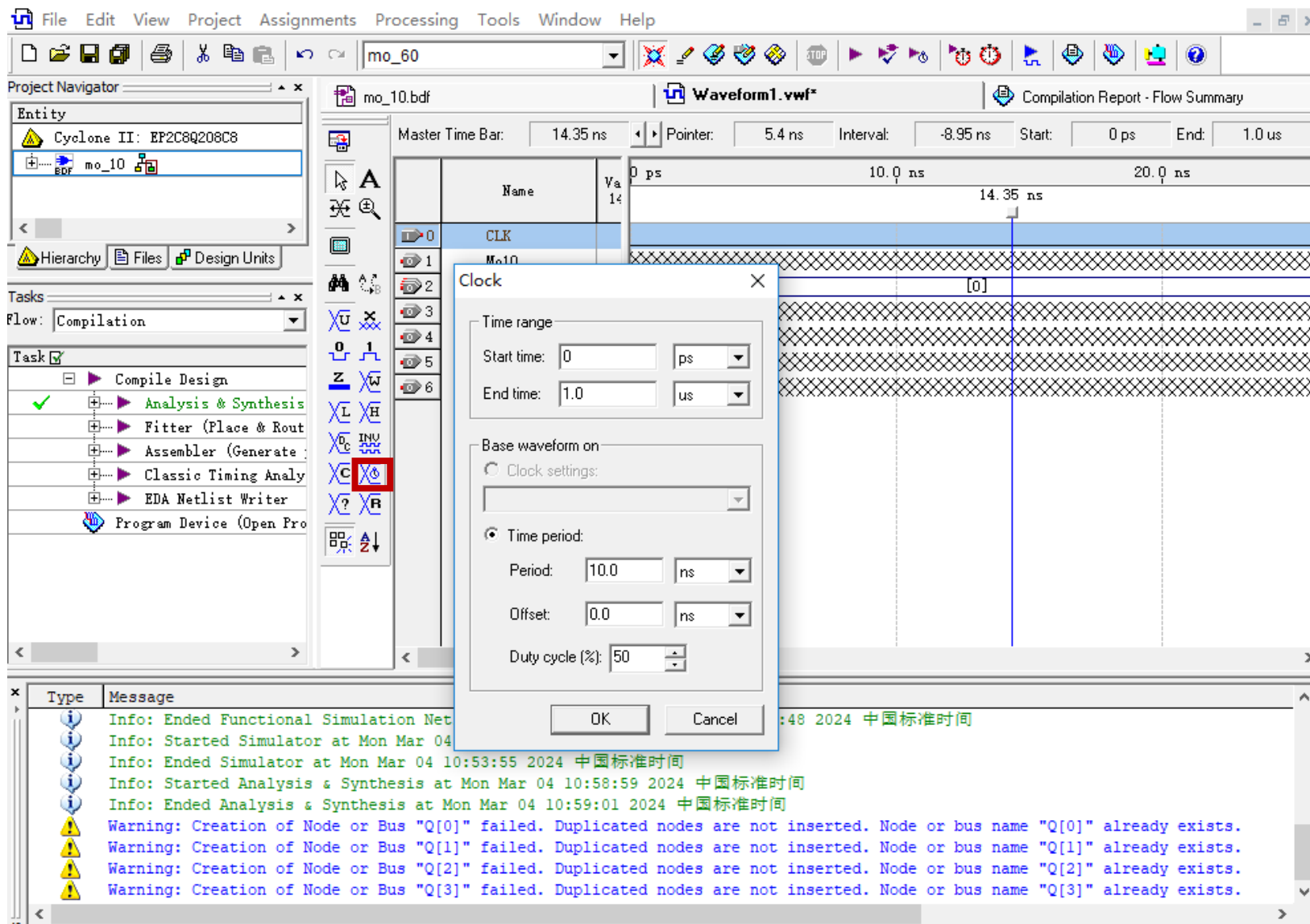
End time extension per signal:

Signal Name	Direction	Radix	Extension value
CLK	Input	ASCII	Default extension value
Mo10	Output	ASCII	Default extension value
+ Q	Output	ASCII	Default extension value

OK Cancel

QuartusII软件的使用— Step4功能仿真

◆ 3、输入信号赋值



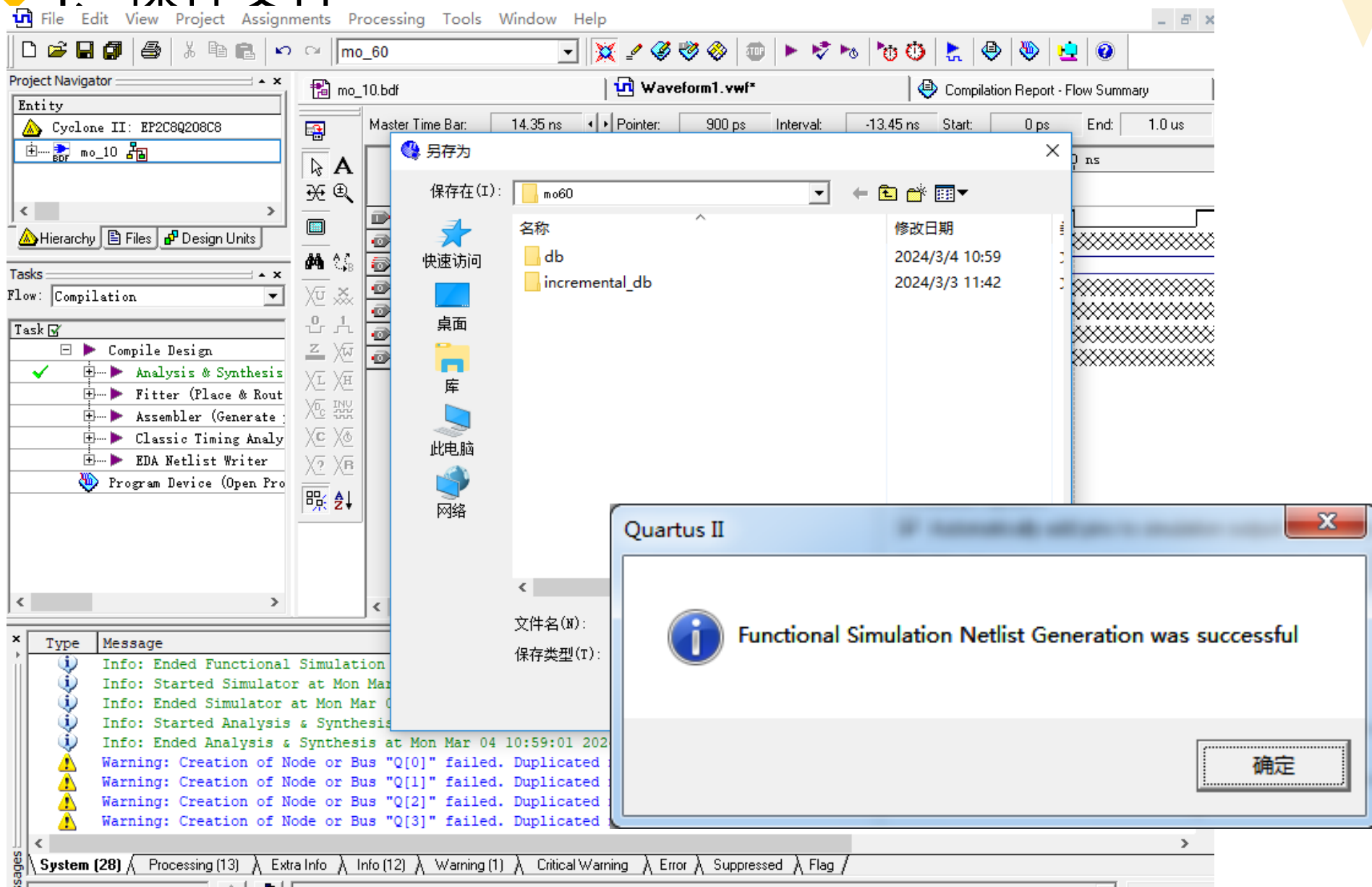
The screenshot displays the Quartus II software interface during a functional simulation setup. The 'Clock' dialog box is open, showing the following configuration:

- Time range:** Start time: 0 ps, End time: 1.0 us.
- Base waveform on:** ☒ Time period.
- Time period settings:** Period: 10.0 ns, Offset: 0.0 ns, Duty cycle (%): 50.

The background shows the Project Navigator with the 'mo_10' component selected, and the Waveform1.vwf window displaying a timing diagram for the CLK signal. The Message window at the bottom shows simulation progress and warnings about duplicated nodes.

Quartus II软件的使用— Step4功能仿真

◆ 4. 保存文件



The screenshot displays the Quartus II software interface. The 'Project Navigator' on the left shows the project hierarchy with 'mo_10' selected. The 'Tasks' pane on the left lists various tasks, including 'Compile Design', 'Analysis & Synthesis', 'Fitter (Place & Rout)', 'Assembler (Generate)', 'Classic Timing Analy', 'EDA Netlist Writer', and 'Program Device (Open Pro)'. The 'Messages' pane at the bottom shows a list of messages, including 'Info: Ended Functional Simulation', 'Info: Started Simulator at Mon Mar 04 10:59:01 2024', 'Info: Ended Simulator at Mon Mar 04 10:59:01 2024', 'Info: Started Analysis & Synthesis at Mon Mar 04 10:59:01 2024', 'Info: Ended Analysis & Synthesis at Mon Mar 04 10:59:01 2024', and several 'Warning: Creation of Node or Bus "Q[0]" failed. Duplicated' messages.

The 'Waveform1.vwf' window is open, showing a waveform plot. The 'Master Time Bar' indicates a time of 14.35 ns, with a pointer at 900 ps, an interval of -13.45 ns, and a start/end range from 0 ps to 1.0 us.

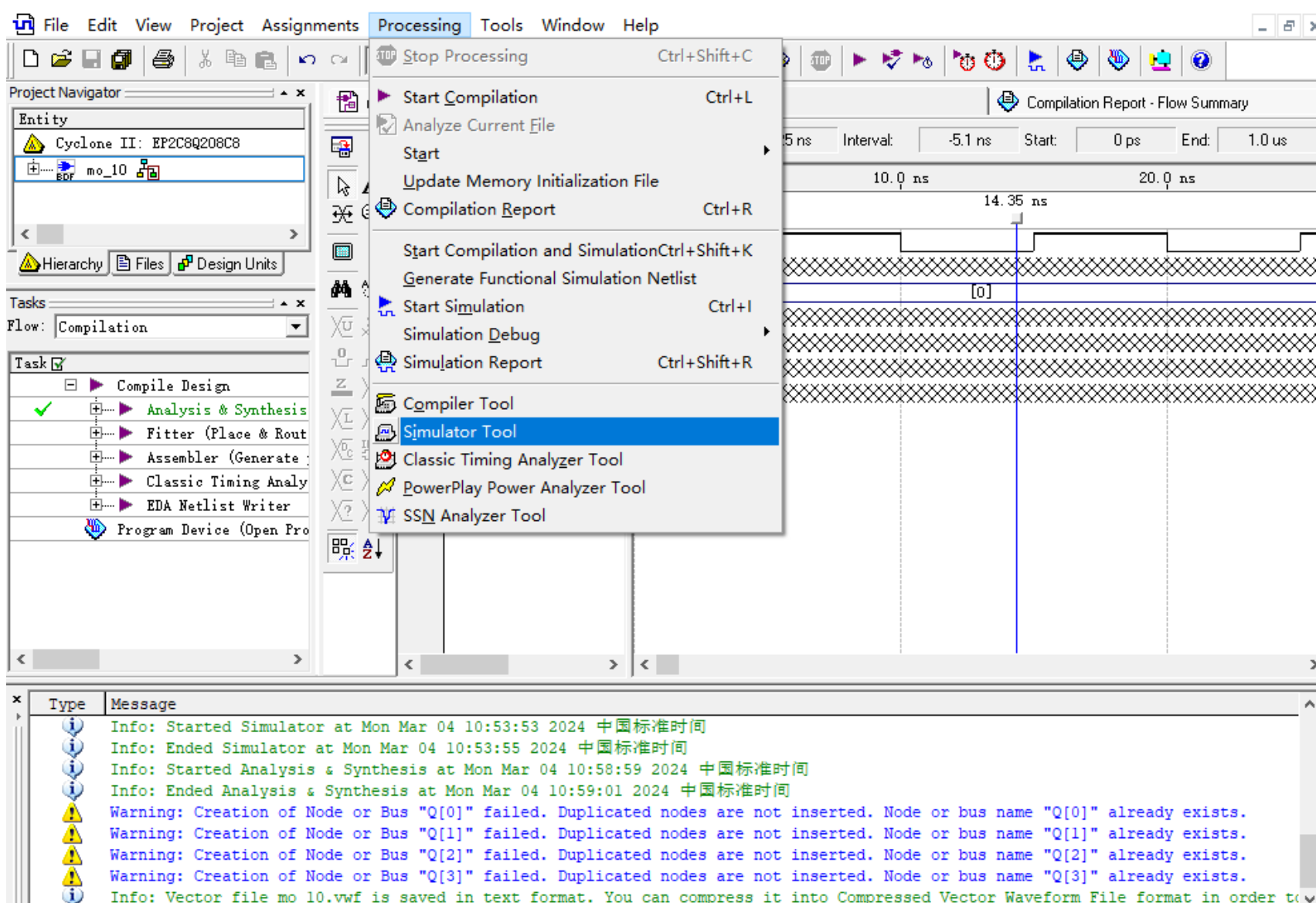
A '另存为' (Save As) dialog box is open, showing the file 'mo_60' being saved in the 'mo60' directory. The dialog lists the following files and their modification dates:

名称	修改日期
db	2024/3/4 10:59
incremental_db	2024/3/3 11:42

The 'Quartus II' message box displays the text: 'Functional Simulation Netlist Generation was successful'. The '确定' (OK) button is visible.

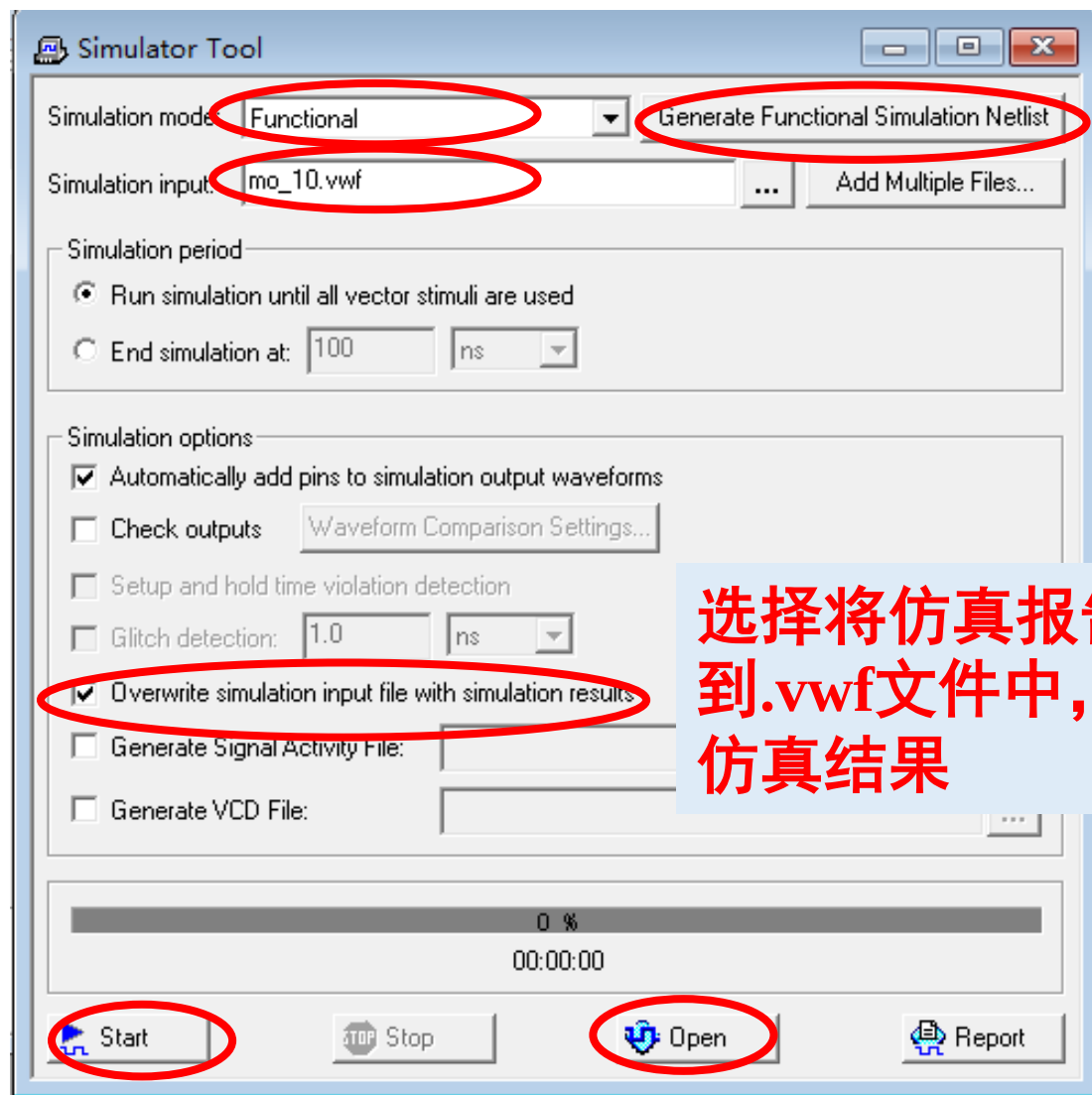
QuartusII软件的使用— Step4功能仿真

◆ 5、功能仿真



QuartusII软件的使用— Step4功能仿真

◆ 5、功能仿真



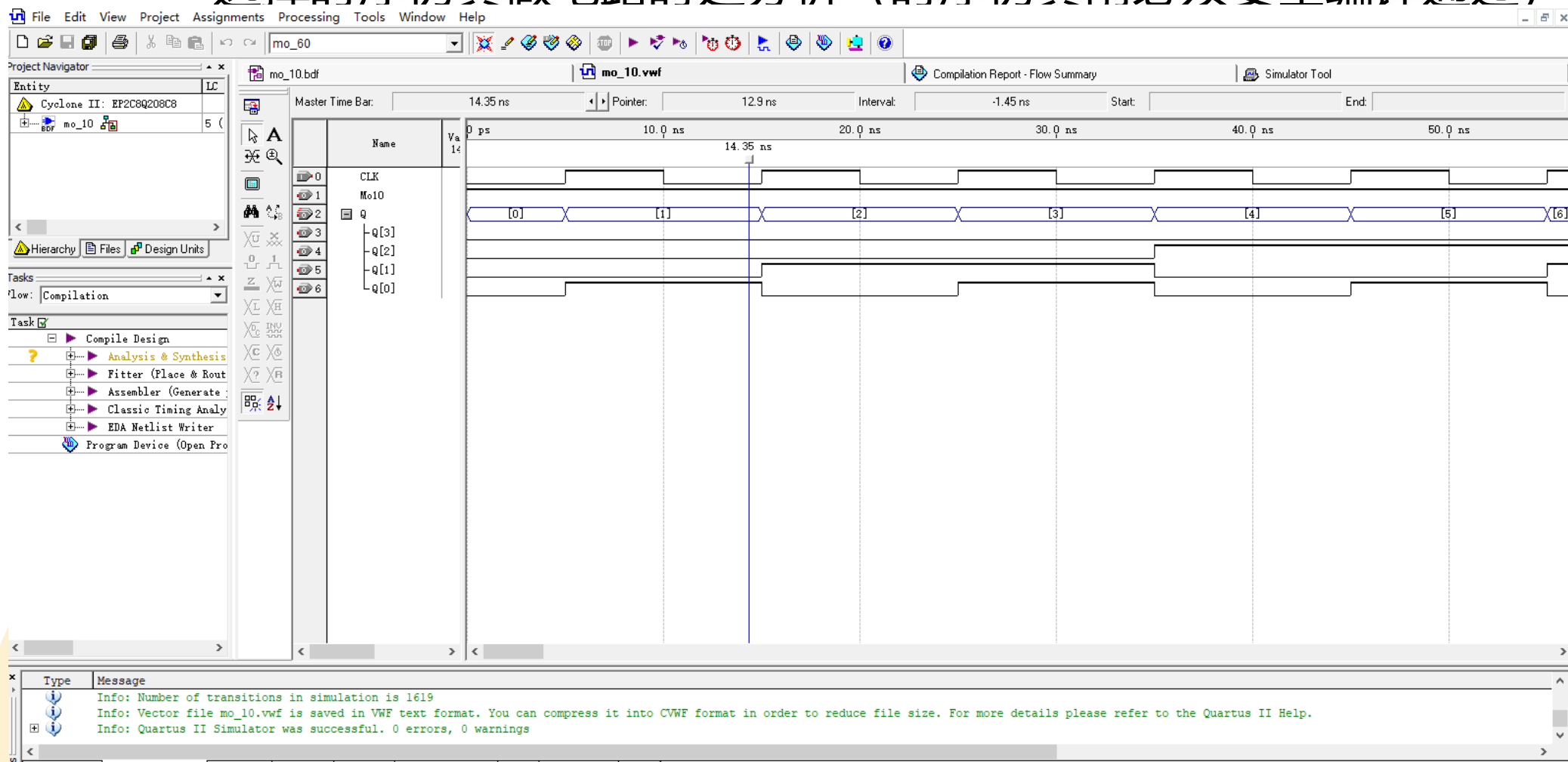
- 选择功能仿真
- 指定波形激励文件
- 生成功能仿真网表文件

选择将仿真报告覆盖到.vwf文件中，可保存仿真结果

QuartusII软件的使用— Step4功能仿真

◆ 5、功能仿真

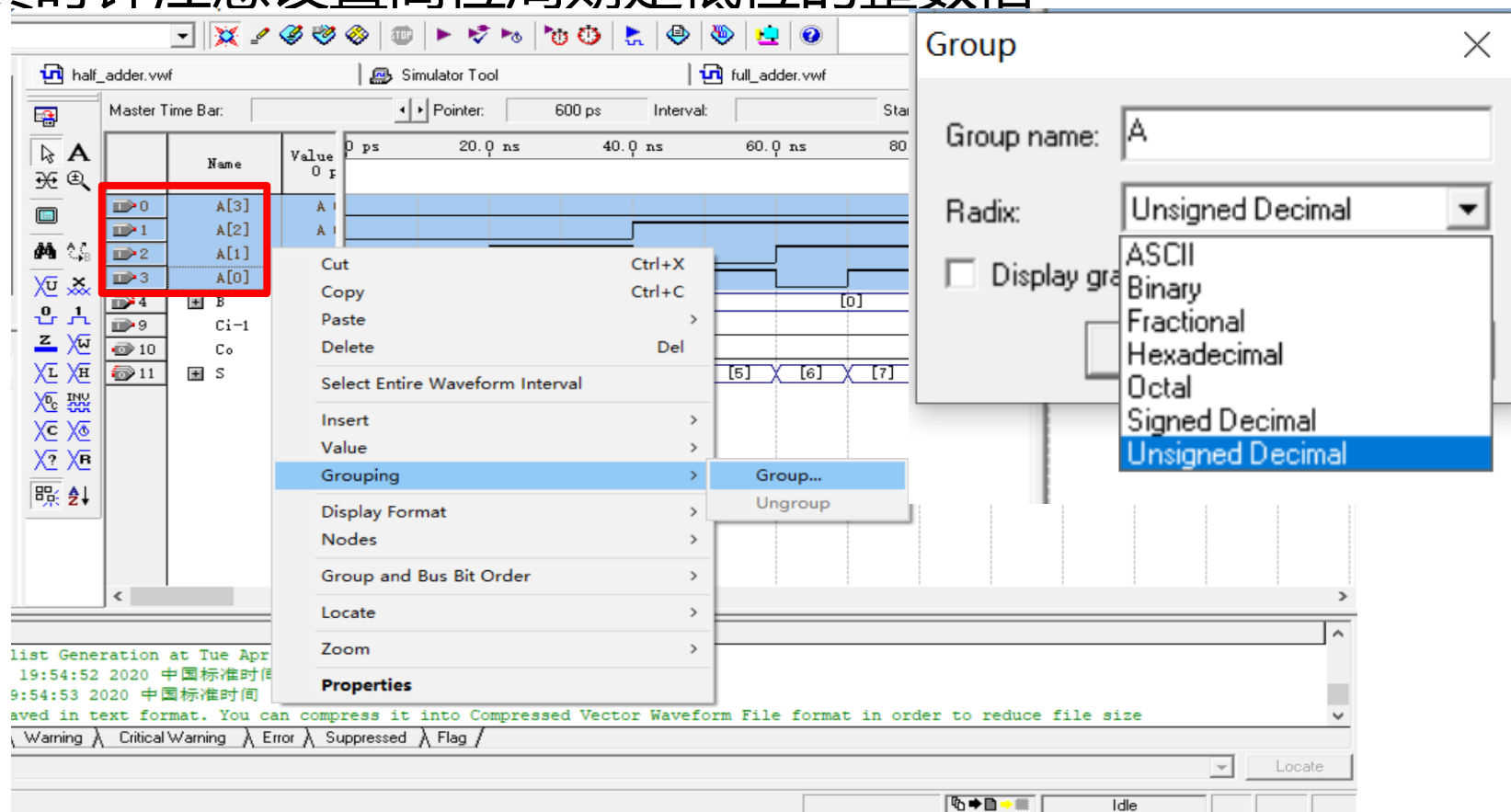
- 功能仿真反映电路功能结果
- 选择时序仿真做电路时延分析（时序仿真前必须要全编译通过）



QuartusII软件的使用— Step4功能仿真

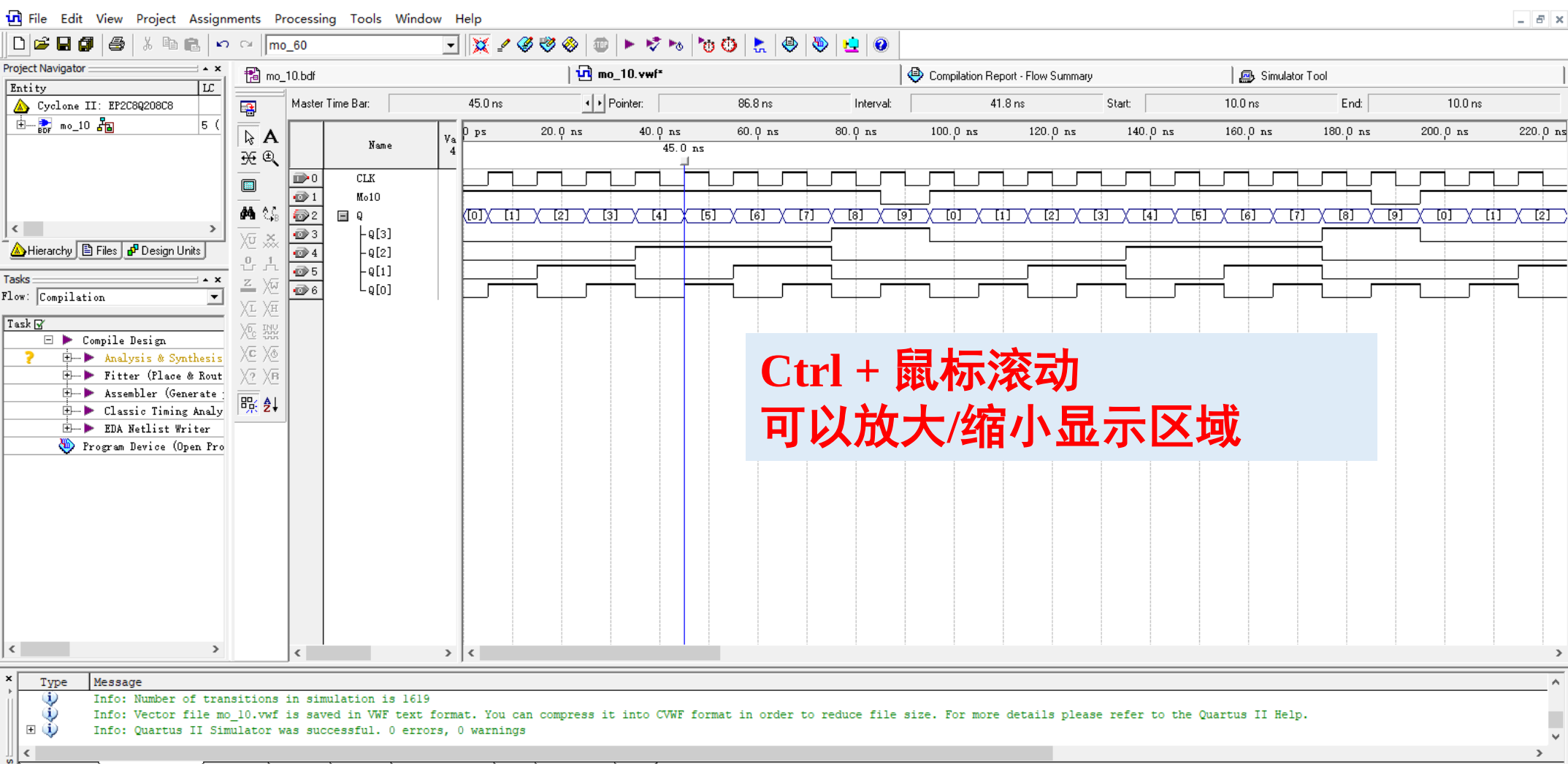
◆ 5、功能仿真

- 仿真结果涉及多位输入/输出时，可将它们从高位到低位的顺序排列好，选择Grouping来成组观测
- 仿真时钟注意设置高位周期是低位的整数倍



QuartusII软件的使用— Step4功能仿真

◆ 5、功能仿真



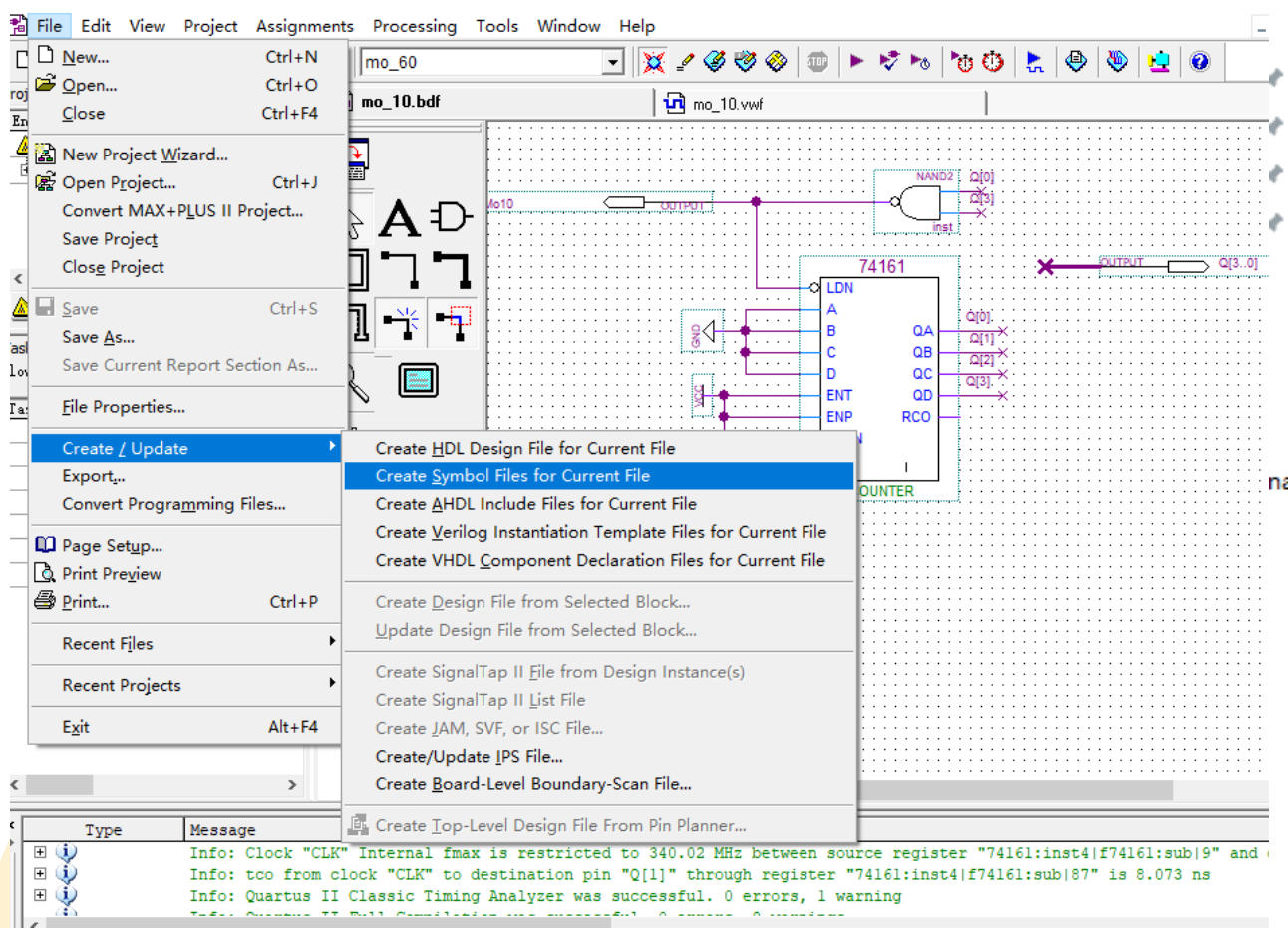
Ctrl + 鼠标滚动
可以放大/缩小显示区域

Type Message
Info: Number of transitions in simulation is 1619
Info: Vector file mo_10.vwf is saved in VWF text format. You can compress it into CVWF format in order to reduce file size. For more details please refer to the Quartus II Help.
Info: Quartus II Simulator was successful. 0 errors, 0 warnings

QuartusII软件的使用— Step5层次化项目设计

◆ 1、封装功能元件为*.bsf图形模块，用于后续引用

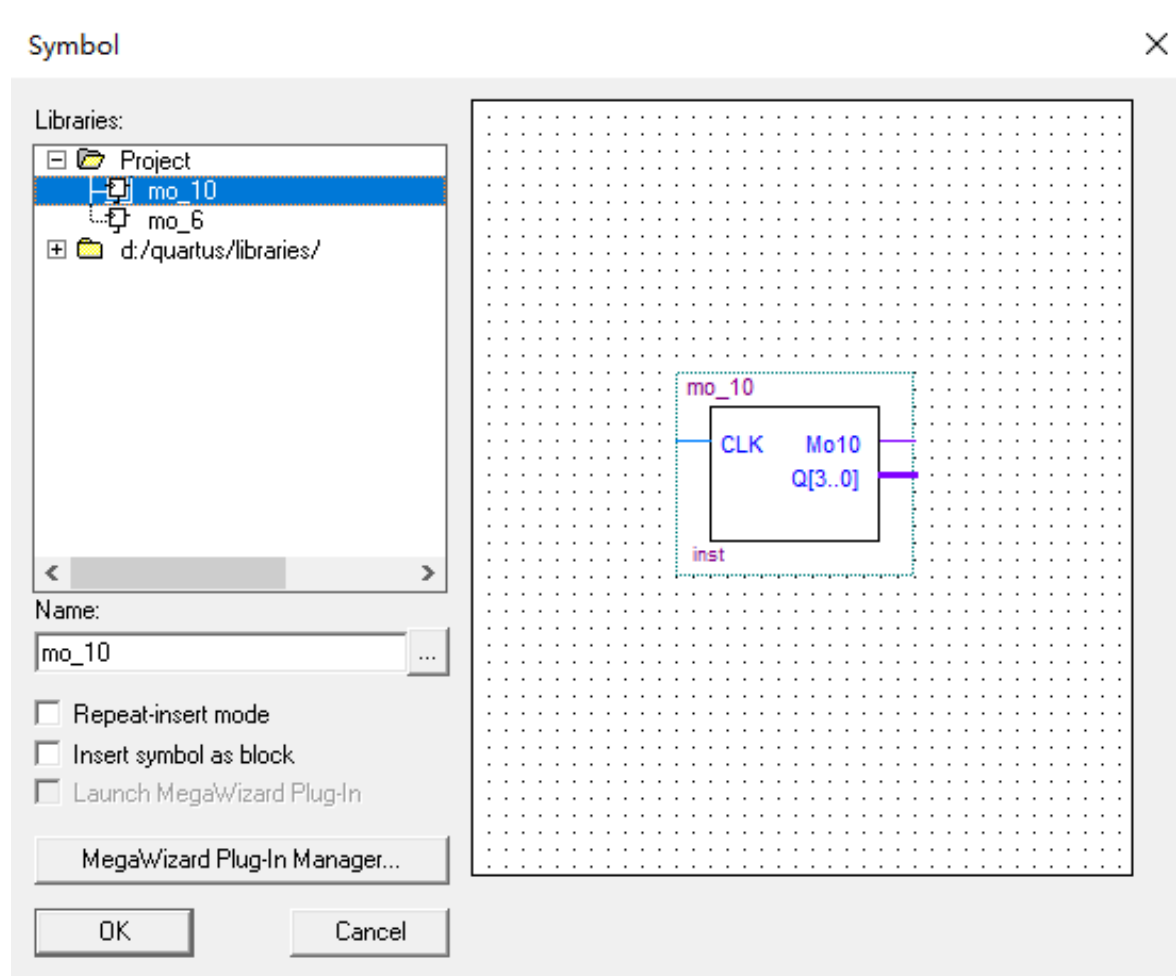
- File -> Create/Update -> Create Symbol File for Current File



名称	类型
db	文件夹
incremental_db	文件夹
mo_10	Quartus II Block/Schematic File
mo_10	Quartus II Block Symbol File
mo_10	Quartus II Vector Waveform File
mo_60.asm.rpt	RPT 文件
mo_60.done	DONE 文件
mo_60.fit.rpt	RPT 文件
mo_60.fit.summary	SUMMARY 文件
mo_60.flow.rpt	RPT 文件
mo_60.map.rpt	RPT 文件
mo_60.map.summary	SUMMARY 文件
mo_60.pin	PIN 文件
mo_60.pof	POF 文件
mo_60	Quartus II Project File
mo_60.qsf	QSF 文件
mo_60.sim.rpt	RPT 文件
mo_60.sof	SOF 文件
mo_60.tan.rpt	RPT 文件
mo_60.tan.summary	SUMMARY 文件

QuartusII软件的使用— Step5层次化项目设计

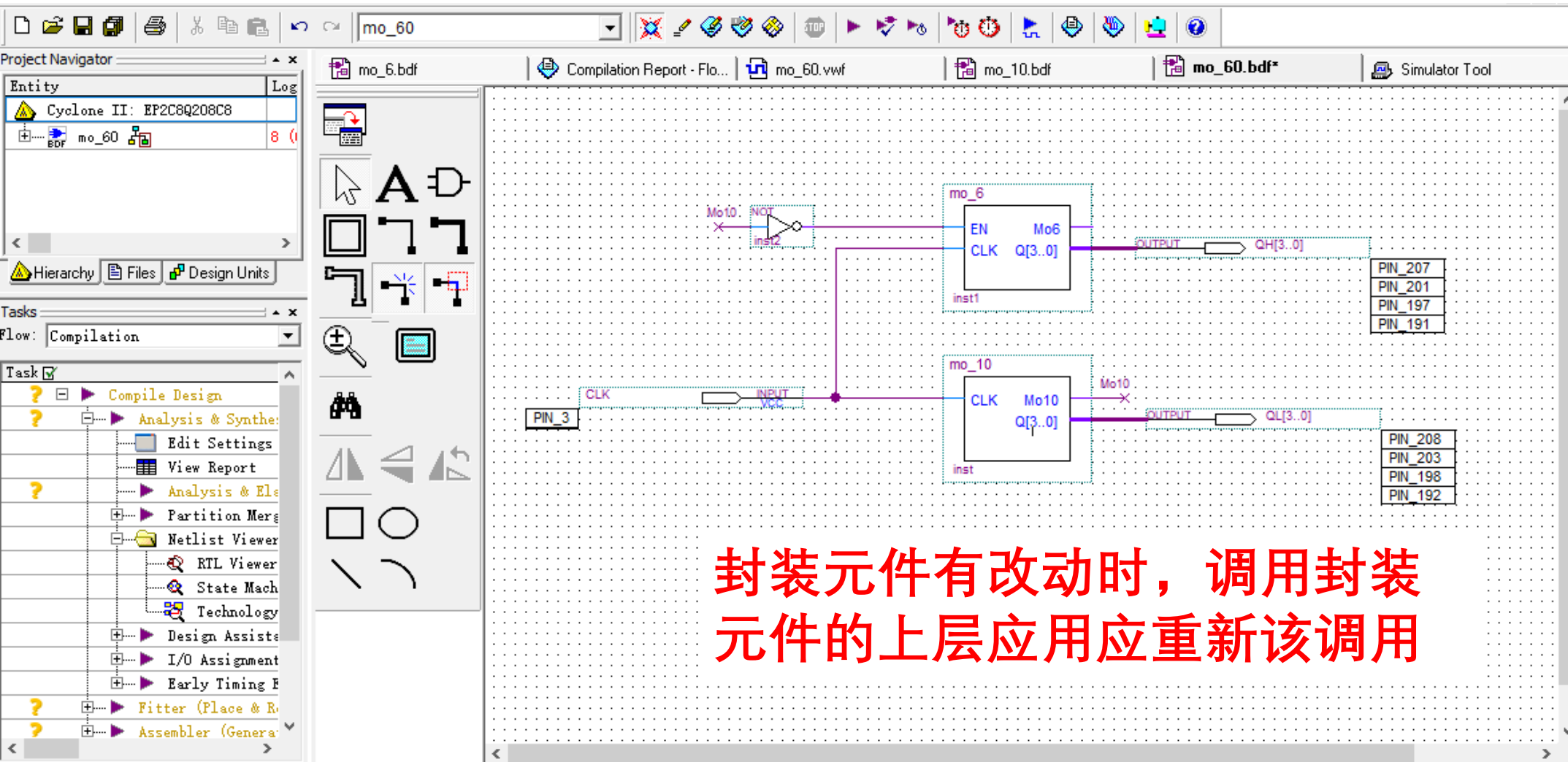
◆ 2、调用封装元件，创建新的原理图





QuartusII软件的使用— Step5层次化项目设计

- ◆ 2、调用封装元件，创建新的原理图
 - 完整模60的设计，并进行功能仿真





Quartus II 软件的使用

— Step6 适配、编程和配置、时序分析

◆ 1、全编译

Flow Summary

Flow Status	Successful - Mon Mar 04 13:26:00 2024
Quartus II Version	9.1 Build 350 03/24/2010 SP 2 SJ Web Edition
Revision Name	mo_60
Top-level Entity Name	mo_60
Family	Cyclone II
EP2C8Q208C8	
Final	Yes
	8 / 8,256 (< 1 %)
	8 / 8,256 (< 1 %)
	7 / 8,256 (< 1 %)
	7
	9 / 138 (7 %)
	0
	0 / 165,888 (0 %)
Embedded Multiplier 9-bit elements	0 / 36 (0 %)
Total PLLs	0 / 2 (0 %)

Quartus II

Full Compilation was successful (7 warnings)

确定

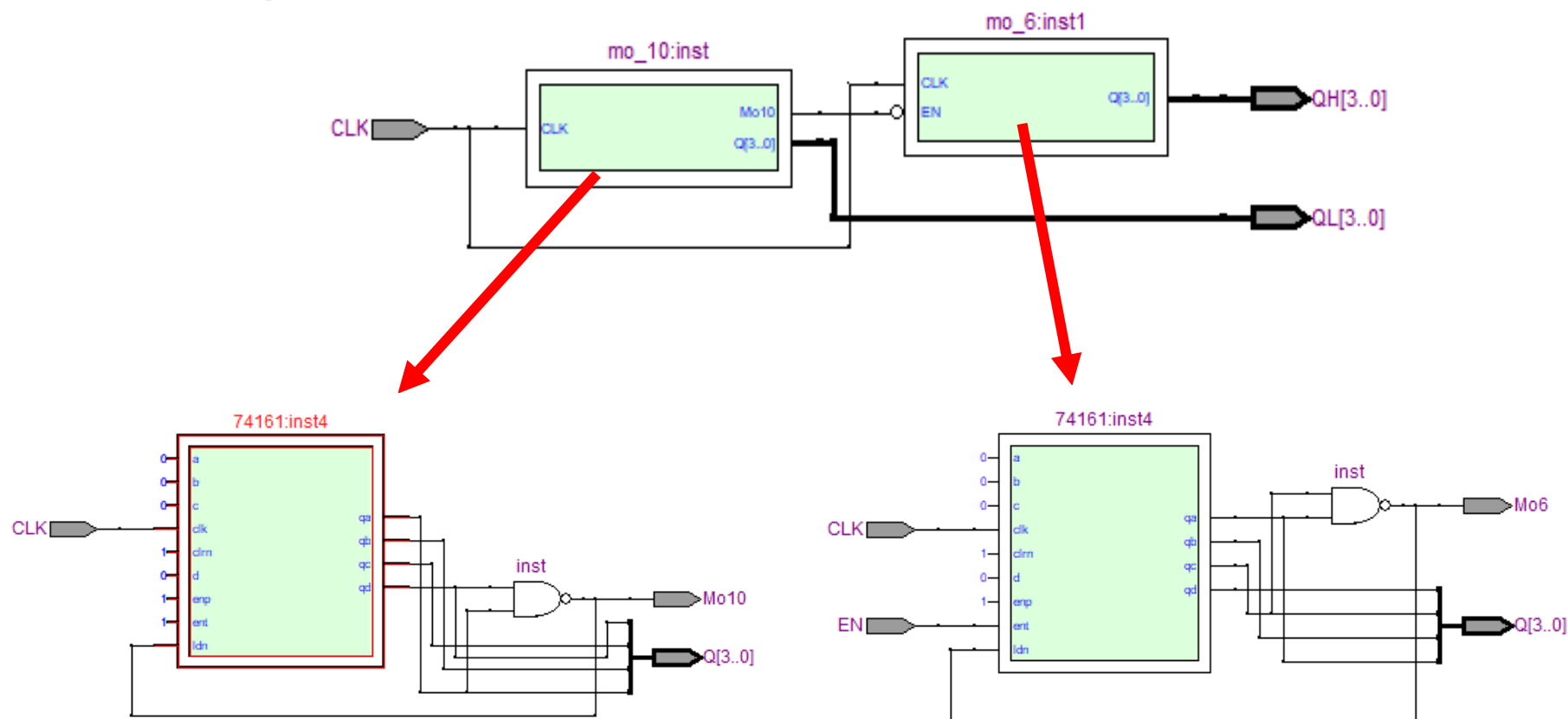
QuartusII软件的使用

— Step6适配、编程和配置、时序分析

数电实验



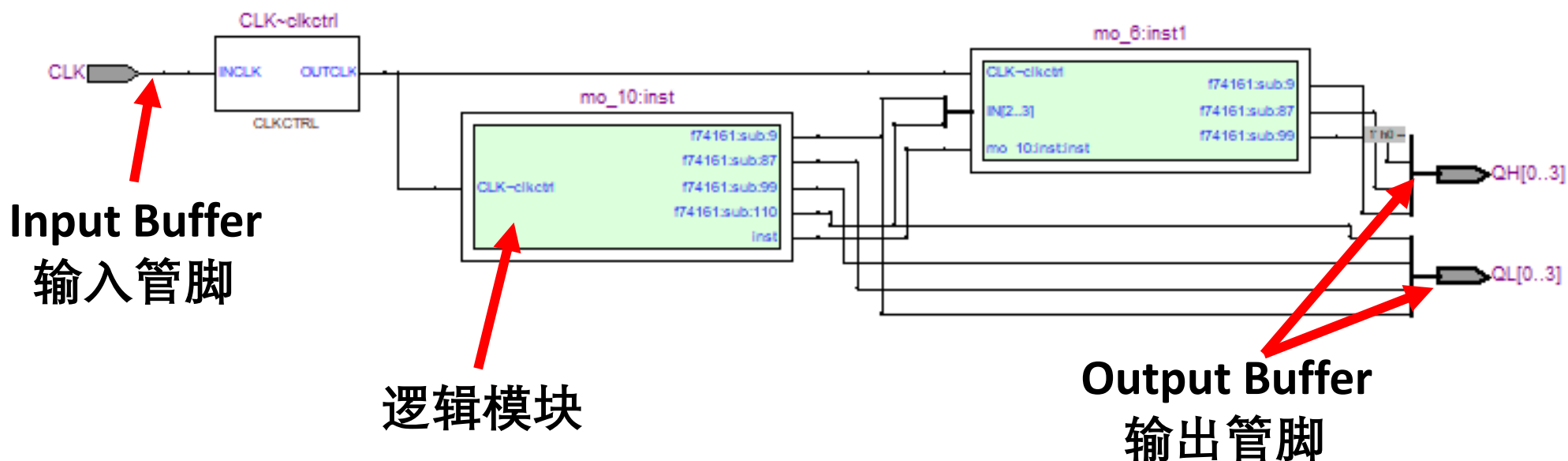
- ◆ 2、用Tools -> Netlist Viewers -> RTL Viewer查看电路综合结果



QuartusII软件的使用

— Step6适配、编程和配置、时序分析

◆ 3、用Tools -> Netlist Viewers -> Technology Map Viewer 查看电路Map结果



- 软件会自动打平设计层次，自动优化逻辑函数
- 设计者要设置好约束条件，无需深度参与

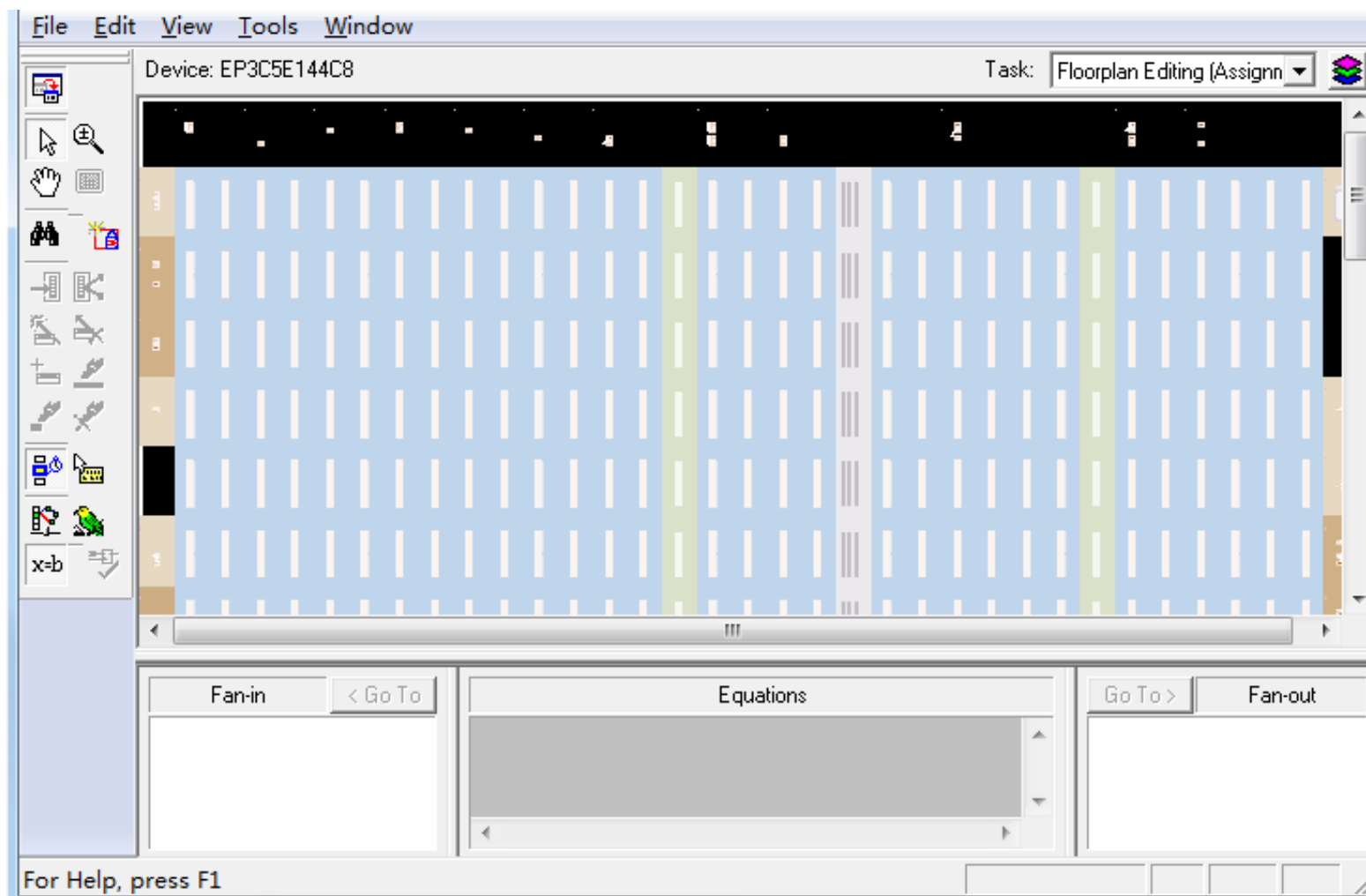
QuartusII软件的使用

— Step6适配、编程和配置、时序分析

数电实验



◆4、Tools -> Chip Planner 查看器件适配结果





— Step6适配、编程和配置、时序分析

◆ 5、时序仿真

- 时序仿真是在功能仿真的基础上利用在布局布线中获得的精确延时参数进行的精确仿真，一般时序仿真的结果和实际结果非常的接近，但由于要计算大量的时延信息，仿真速度比较慢。
- 时序仿真前需进行全编译

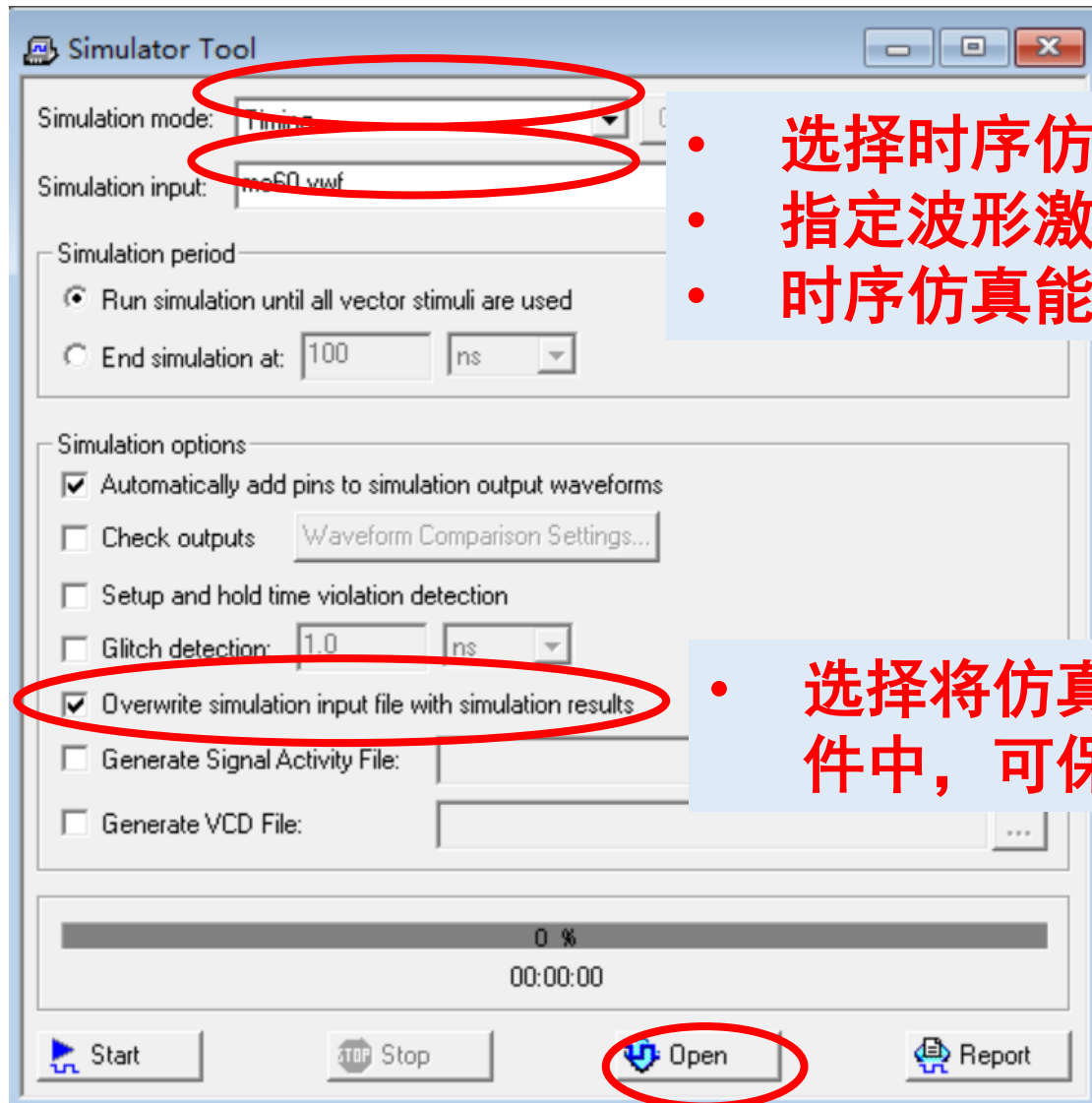
QuartusII软件的使用

— Step6适配、编程和配置、时序分析

数电实验



◆ 5、时序仿真 Processing -> Simulator Tool



- 选择时序仿真
- 指定波形激励文件
- 时序仿真能反映出硬件延迟的影响

- 选择将仿真报告覆盖到.vwf文件中，可保存仿真结果



竞争和冒险

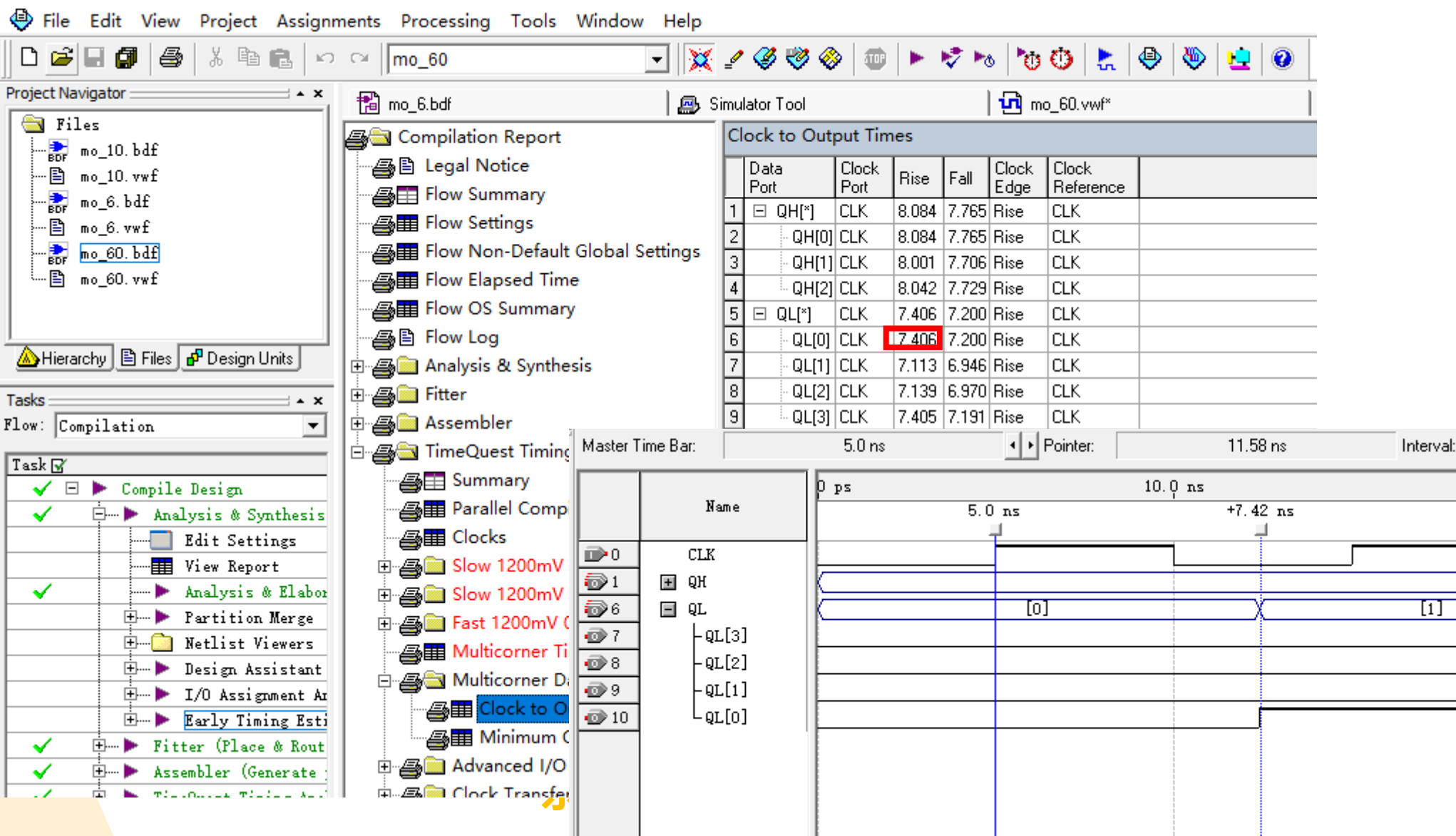
QuartusII软件的使用

— Step6适配、编程和配置、时序分析

数电实验



◆5、时序仿真



分配管脚

- ◆ Analysis and Synthesis全部通过后，为了把的设计下载到实际电路中进行验证，还必须把设计项目的输入输出端口和器件相应的引脚绑定在一起
- ◆ 分配管脚
 - 给引脚分配信号
 - 给信号分配管脚
- ◆ Assignments -> Pins，为所有的I/O配置管脚

QuartusII软件的使用— Step7分配管脚

◆ FPGA实验板

- 查表确认FPGA接口引脚编号
- 区分通用IO口及功能IO口
- 安装usb-blaster驱动，确认设备连接电脑

J6-IO扩展接口

USB口数据&供电

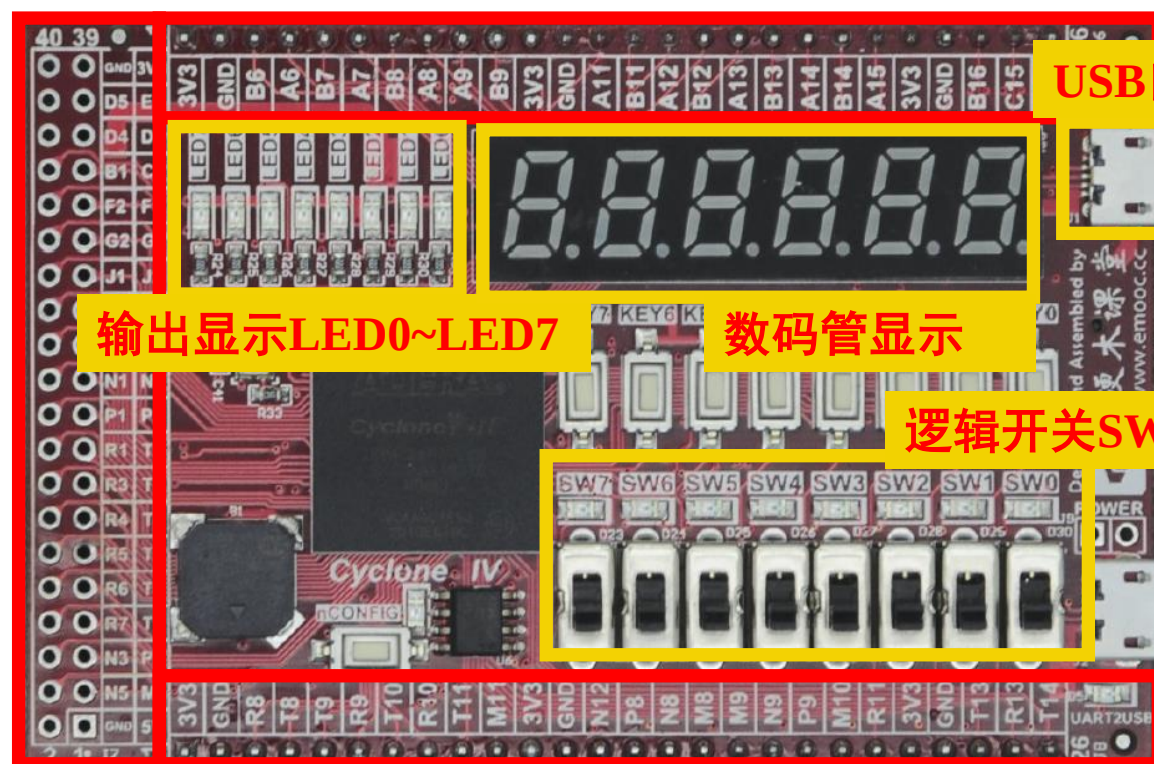
输出显示LED0~LED7

数码管显示

逻辑开关SW0~SW7

J8-IO扩展接口

东南大学电工电子实验中心



J7-IO
扩展
接口

QuartusII软件的使用— Step7分配管脚

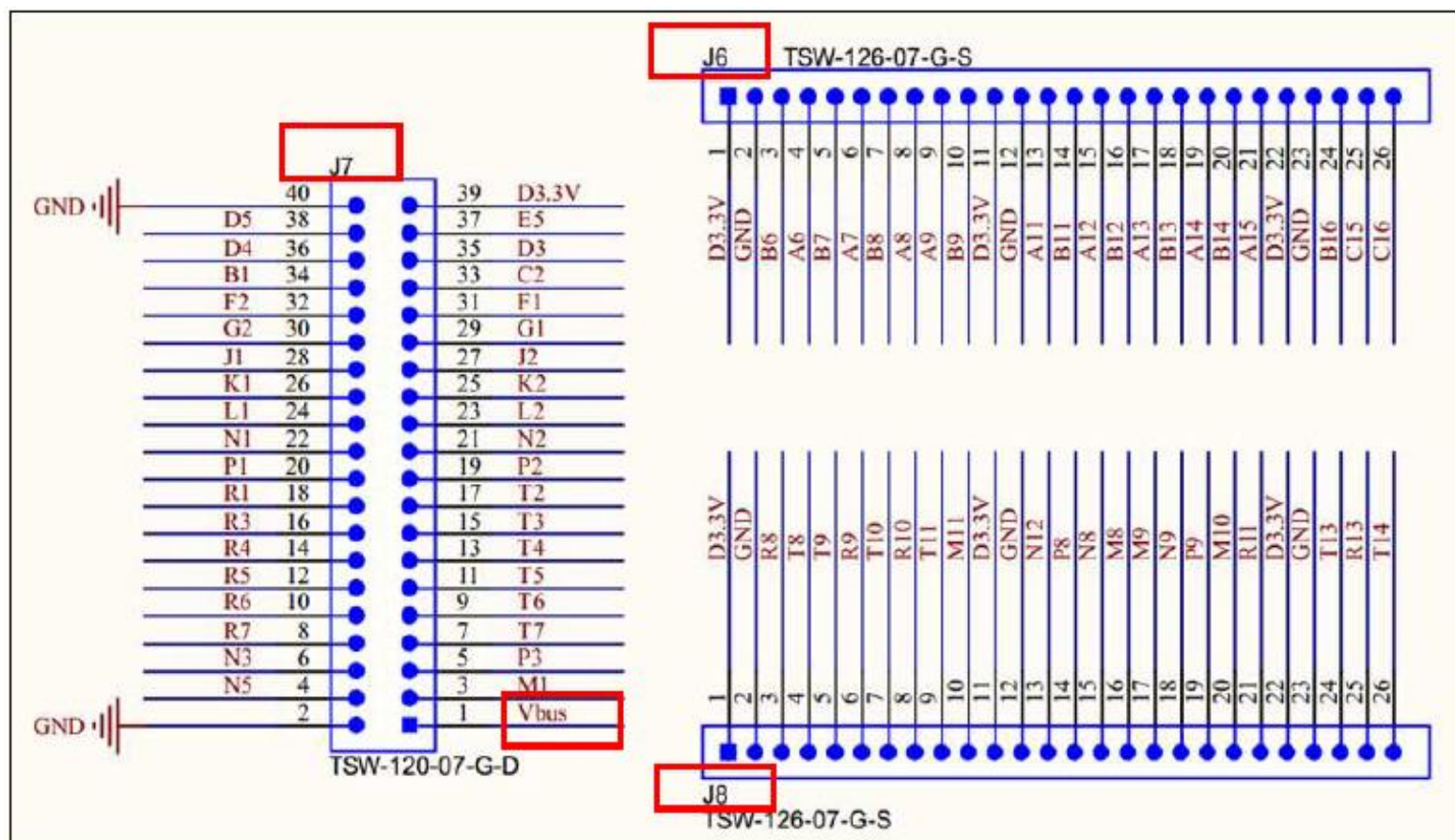
◆FPGA实验板

拨动开关引脚配置			LED 引脚配置		
信号名	FPGA引脚号	说明	信号名	FPGA引脚号	说明
SW_In[0]	PIN_R16	SW0	LED_Out[0]	PIN_A5	LED0
SW_In[1]	PIN_P15	SW1	LED_Out[1]	PIN_B5	LED1
SW_In[2]	PIN_P16	SW2	LED_Out[2]	PIN_A4	LED2
SW_In[3]	PIN_N15	SW3	LED_Out[3]	PIN_B4	LED3
SW_In[4]	PIN_N16	SW4	LED_Out[4]	PIN_A3	LED4
SW_In[5]	PIN_M12	SW5	LED_Out[5]	PIN_B3	LED5
SW_In[6]	PIN_N14	SW6	LED_Out[6]	PIN_A2	LED6
SW_In[7]	PIN_N13	SW7	LED_Out[7]	PIN_C3	LED7

QuartusII软件的使用— Step7分配管脚

◆FPGA实验板

- 40 引脚的IO 扩展接口模块J7 和两个26 引脚的IO 接口模块J6、 J8
- 每个连接器引脚旁都标注有对应的FPGA引脚号



QuartusII软件的使用— Step7分配管脚

◆ Assignments -> Device 切换装置，选择Cyclone IV E -EP4CE6F17C8

Settings - mo_60

Category:

- General
- Files
- Libraries
- Device
- Operating Settings and Conditions
- Compilation Process Settings
- EDA Tool Settings
- Analysis & Synthesis Settings
- Filter Settings
- Timing Analysis Settings
- Assembler
- Design Assistant
- SignalTap II Logic Analyzer
- Logic Analyzer Interface
- Simulator Settings
- PowerPlay Power Analyzer Settings
- SSN Analyzer

Device

Select the family and device you want to target for compilation.

Device family:
Family: Cyclone IV E

Devices: All

Target device:
☐ Auto device selected by the Filter
☒ Specific device selected in 'Available devices' list
☐ Other: n/a

Available devices:

Name	Core v...	LEs	User I/O...	Memor...	Embed...	PLL
EP4CE6E22I7	1.2V	6272	92	276480	30	2
EP4CE6E22I8L	1.0V	6272	92	276480	30	2
EP4CE6F17A7	1.2V	6272	180	276480	30	2
EP4CE6F17C6	1.2V	6272	180	276480	30	2
EP4CE6F17C7	1.2V	6272	180	276480	30	2
EP4CE6F17C8	1.2V	6272	180	276480	30	2
EP4CE6F17C8L	1.0V	6272	180	276480	30	2

Migration compatibility:
Migration Devices...
0 migration devices selected

Companion device:
HardCopy:
☒ Limit DSP & RAM to HardCopy device resources

OK Cancel

QuartusII软件的使用— Step7分配管脚

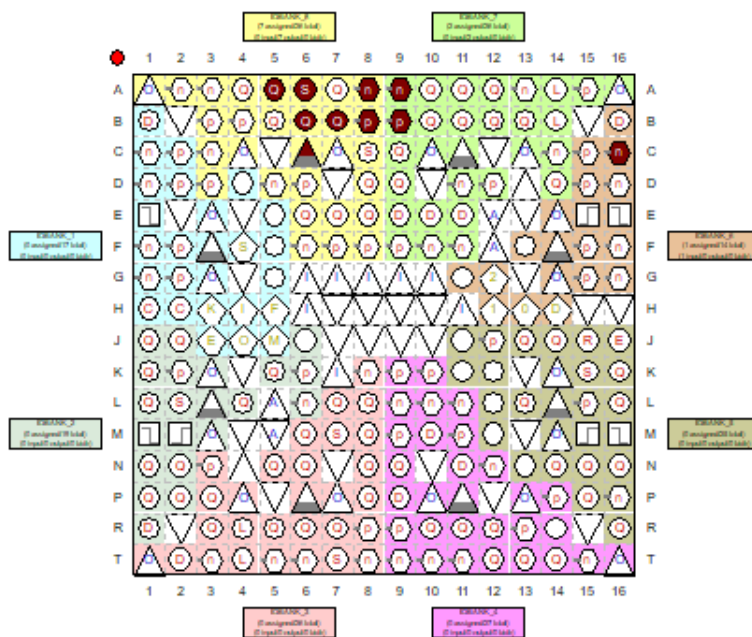
File Edit View Processing Tools Window

Groups

Named: []

Node Name	Direction
QH[3..0]	Output
QL[3..0]	Output
<<new node>>	

Top View - Wire Bond
Cyclone IV E - EP4CE6F17C8

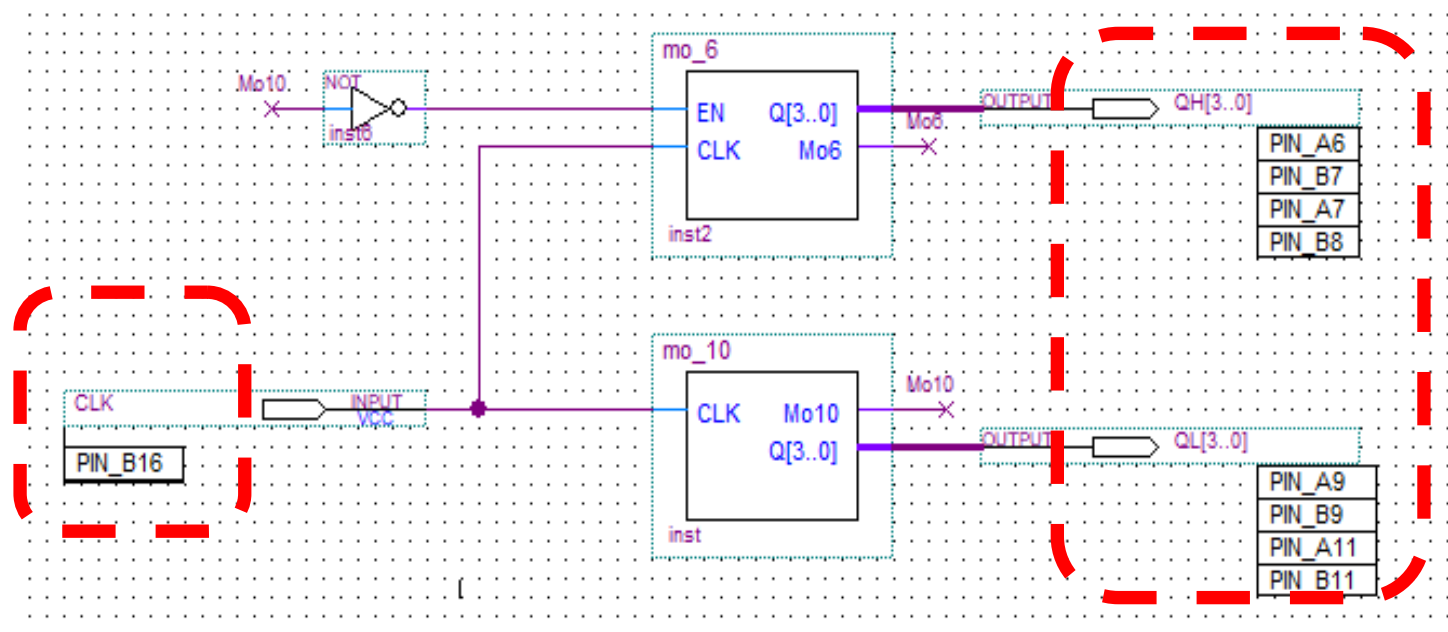


Named: [] Edit: [X] [✓] Filter: Pins: all

	Node Name	Direction	Location	I/O Bank	VREF Group	I/O Standard	Reserved
1	CLK	Input	PIN_C16	6	B6_N0	3.3-V LVTTTL (default)	
2	Mo60	Output	PIN_A5	8	B8_N0	3.3-V LVTTTL (default)	
3	QH[3]	Output	PIN_C6	8	B8_N0	3.3-V LVTTTL (default)	
4	QH[2]	Output	PIN_B6	8	B8_N0	3.3-V LVTTTL (default)	
5	QH[1]	Output	PIN_A6	8	B8_N0	3.3-V LVTTTL (default)	
6	QH[0]	Output	PIN_B7	8	B8_N0	3.3-V LVTTTL (default)	
7	QL[3]	Output	PIN_B8	8	B8_N0	3.3-V LVTTTL (default)	
8	QL[2]	Output	PIN_A8	8	B8_N0	3.3-V LVTTTL (default)	
9	QL[1]	Output	PIN_A9	7	B7_N0	3.3-V LVTTTL (default)	
10	QL[0]	Output	PIN_B9	7	B7_N0	3.3-V LVTTTL (default)	
11	<<new node>>						

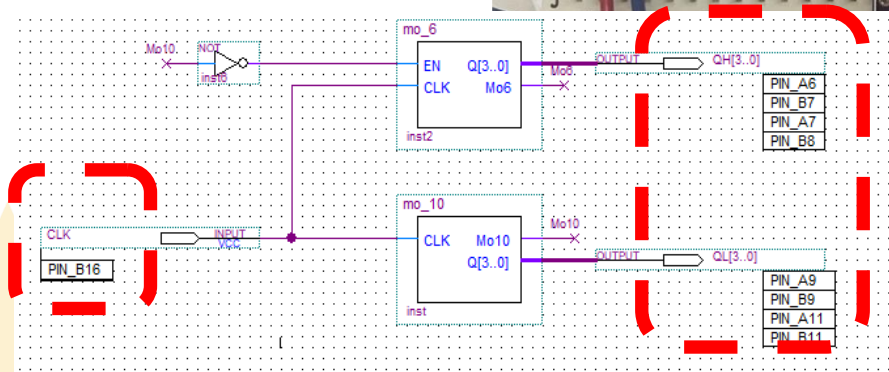
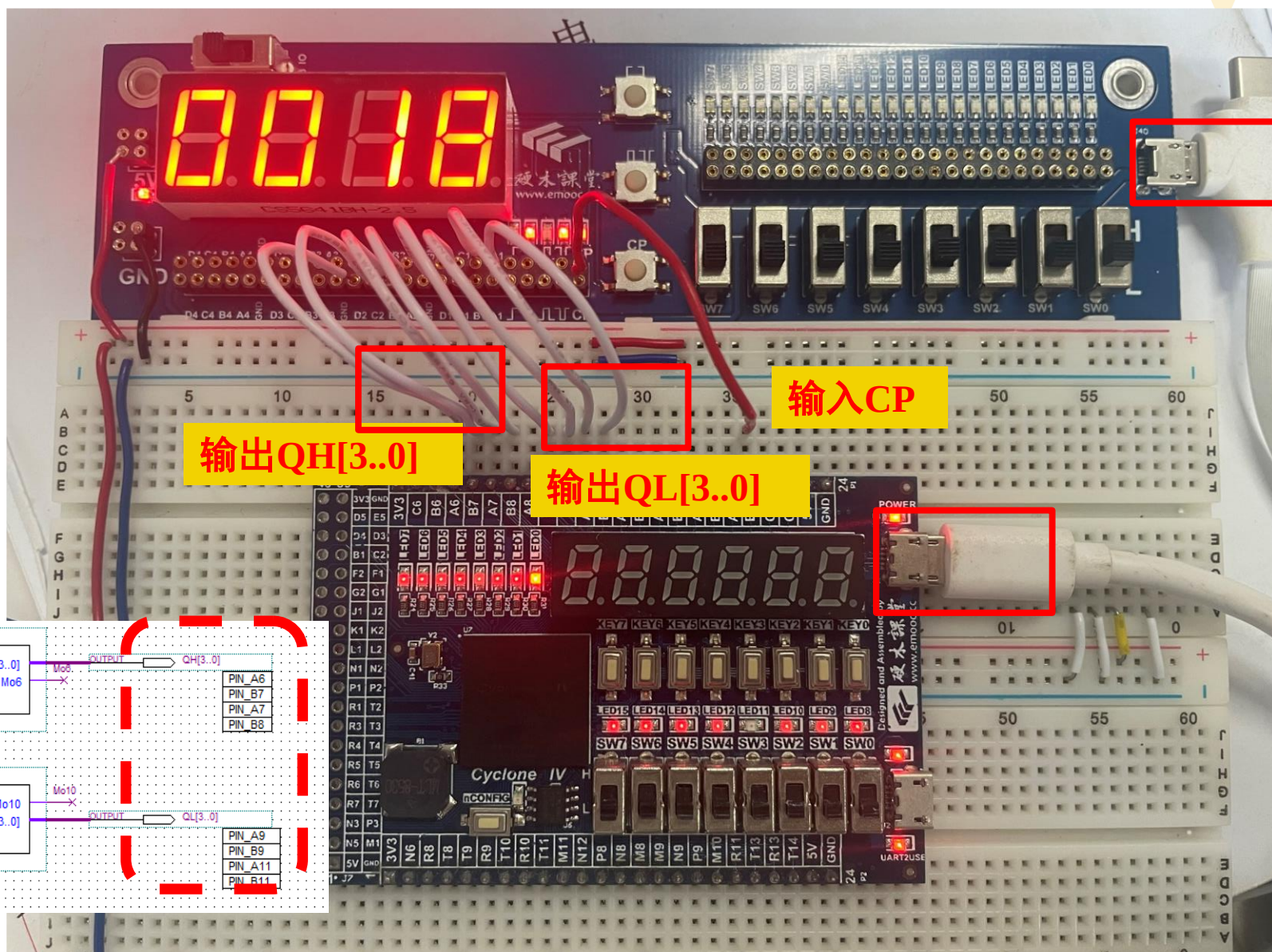
QuartusII软件的使用— Step7分配管脚

◆ Assignments -> Pins , 为所有的I/O配置管脚



QuartusII软件的使用— Step7分配管脚

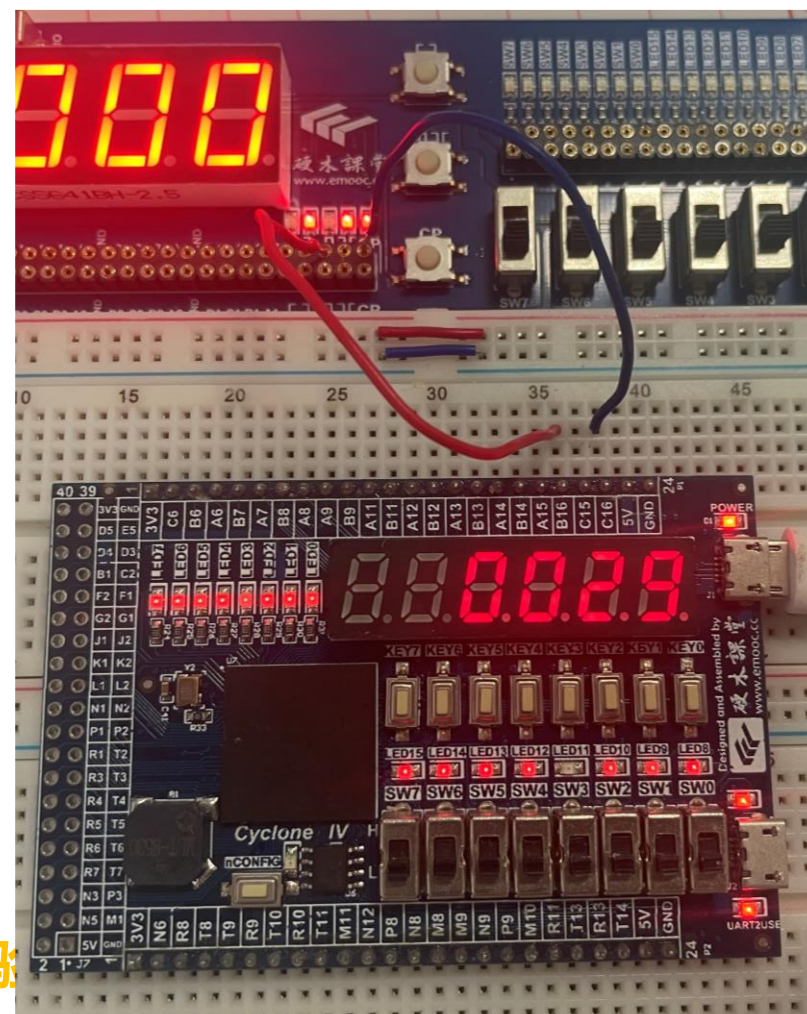
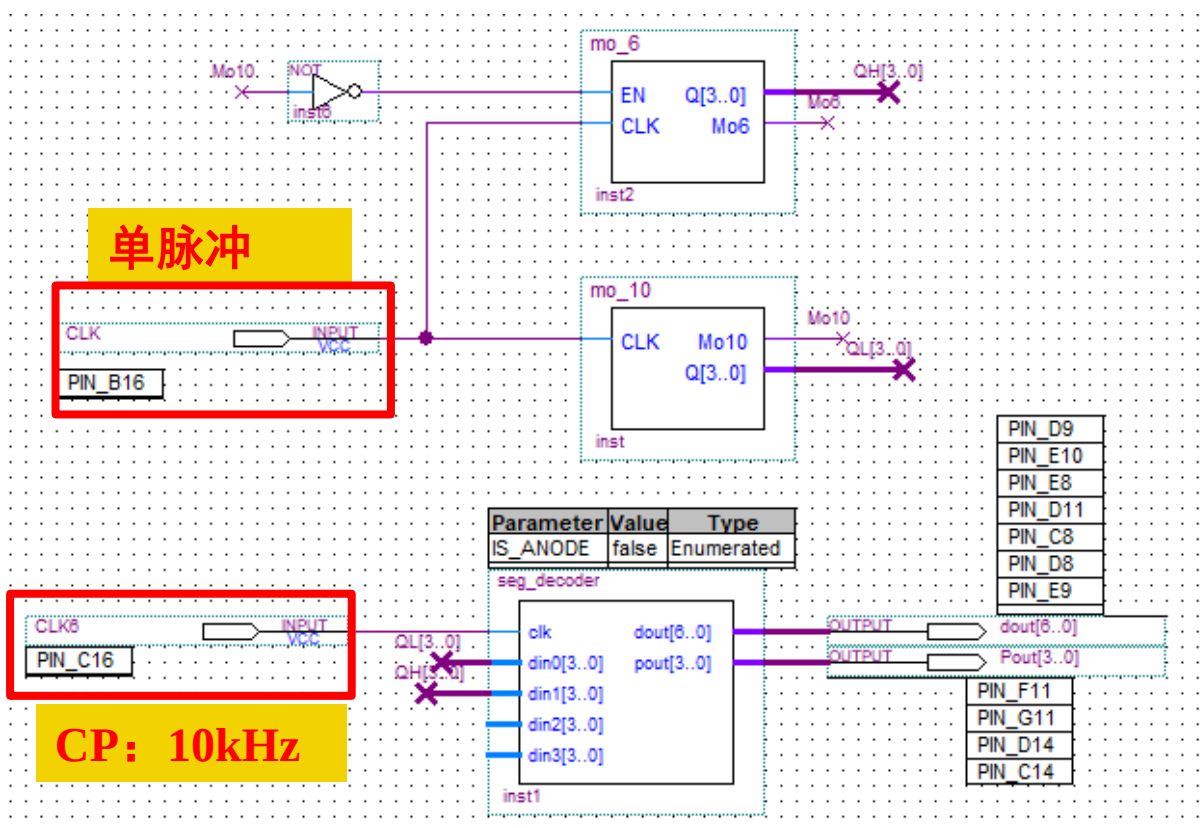
◆显示方法1：使用TB板数码管显示



QuartusII软件的使用— Step7分配管脚

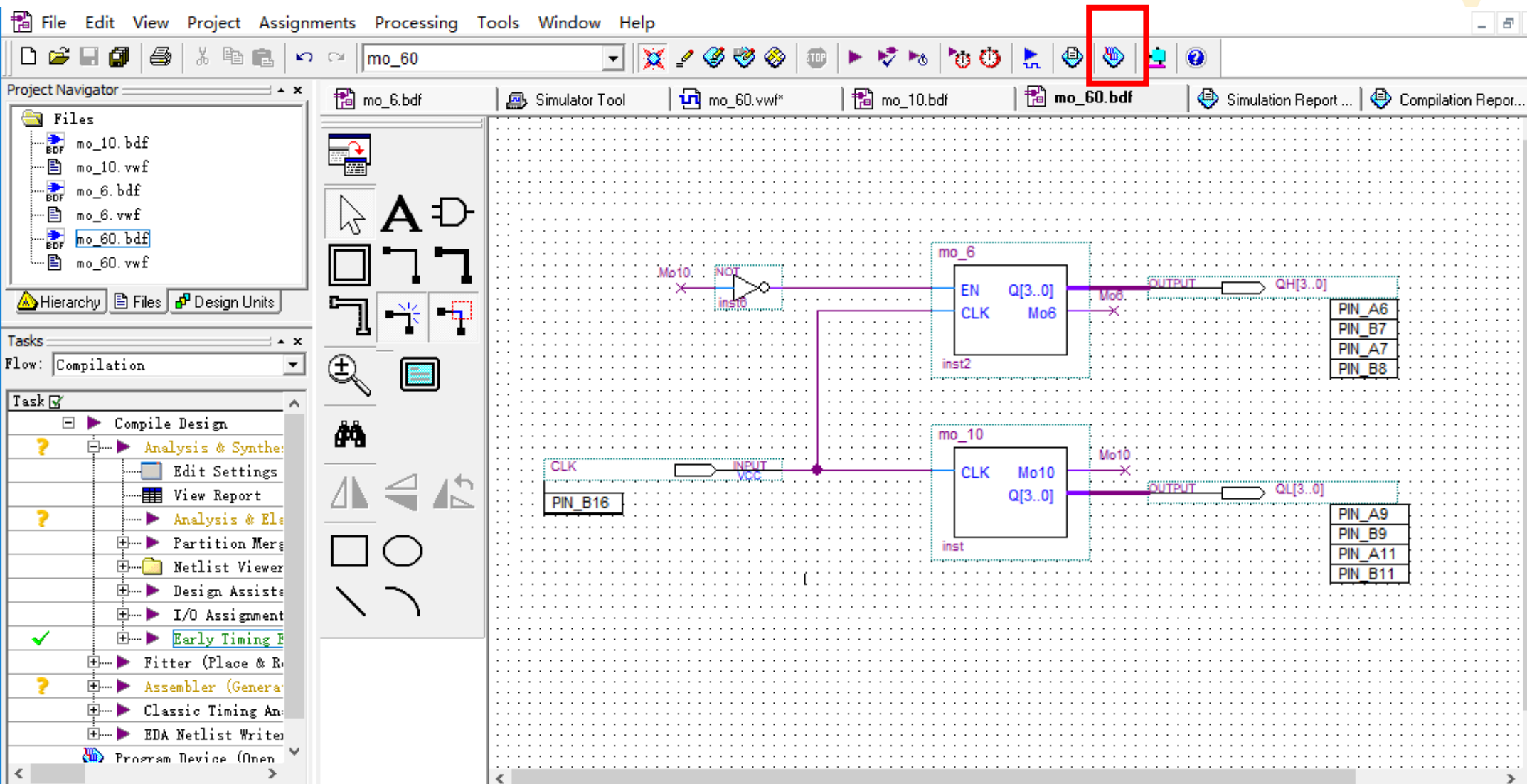
◆显示方法2：使用FPGA数码管显示

- 添加seg_decoder模块，并给该模块分配管脚
(详见数电C-FPGA简要说明)



QuartusII软件的使用— Step8 下载验证

◆连接下载电缆，选择Tools → Programmer菜单



QuartusII软件的使用— Step8 下载验证

◆ 连接下载电缆，选择Tools → Programmer菜单

File Edit Processing Tools Window

Hardware Setup... USB-Blaster [USB-0] Mode: JTAG

☐ Enable real-time ISP to allow background programming (for MAX II devices)

File	Device	Checksum	Usercode	Program/Configure	Verify	Blank-Check	Examine	Security Bit	Erase	ISP CLAMP
mo_60.sof	EP4CE6F17	000887FA	FFFFFFF	☒	☐	☐	☐	☐	☐	☐

Start Stop Auto Detect Delete Add File... Change File... Save File... Add Device... Up Down

Hardware Setup

Hardware Settings JTAG Settings

Select a programming hardware setup to use when programming devices. This programming hardware setup applies only to the current programmer window.

Currently selected hardware: USB-Blaster [USB-0]

Available hardware items:

Hardware	Server	Port
USB-Blaster	Local	USB-0

Add Hardware... Remove Hardware

双击选定

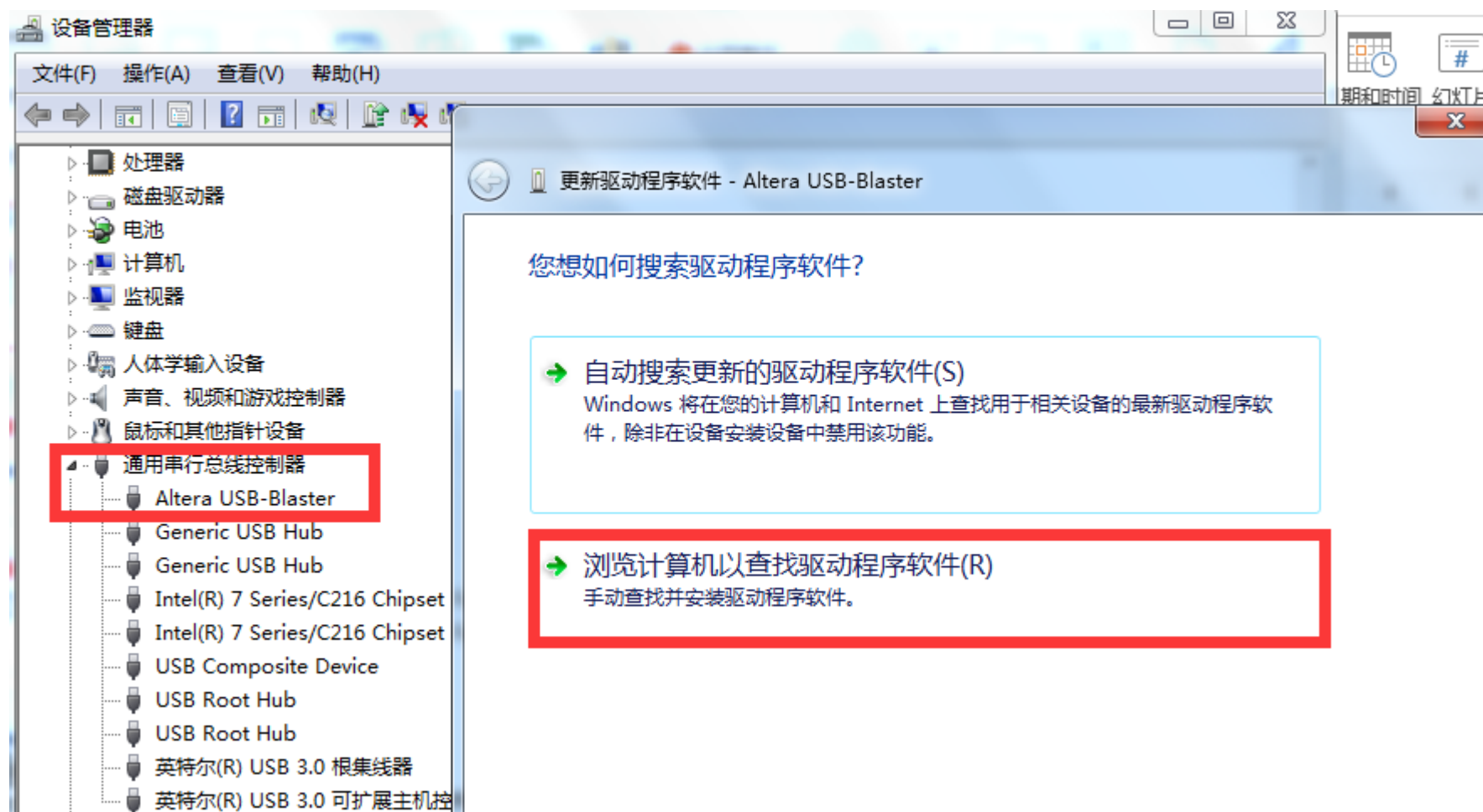
For Help, press F1

单击此处添加备注

Close

QuartusII软件的使用— Step8 下载验证

◆硬件驱动的安装（见教学计划）



QuartusII软件的使用— Step8 下载验证

下载到FPGA，进行验证

编程模式选择

Quartus II - E:/quartus_lhy/mo60/mo_60 - mo_60 - [mo_60.cdf]

File Edit Processing Tools Window

Hardware Setup... USB-Blaster [USB-0] Mode: JTAG Progress: 100 %

☐ Enable real-time ISP to allow background programming (for MAX II devices)

Start Stop Auto Detect Delete Add File... Change File... Save File... Add Device... Up Down

File	Device	Checksum	Usercode	Program/Configure	Verify	Blank-Check	Examine	Security Bit	Erase	ISP CLAMP
mo_60.sof	EP4CE6F17	000887FA	FFFFFFFF	☒	☐	☐	☐	☐	☐	☐

编程文件

编程进度条



实验验收

◆ 模60下载验证



下节课注意事项

- ◆学习慕课第五、第六章相关内容
- ◆课前完成“交通灯实验控制电路”的电路设计，将设计方案写在实验报告的原理部分，并用 Quartus 原理图输入法完成必做内容的原理图绘制
 - 实验电路要求在进实验室前完成设计，实验室内以下载验证为主
- ◆上课前交实验六内容2 “交通灯实验控制电路”的预习报告



Thank you!